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Data Engineering

DIMENSIONAL MODELING – DATA INTEGRATION

Steve Hillen

Introduction

Steve Hillen

- Former student IMS/DSPS/ADI (1998-2001)
- Founder of Agiliz

Evaluation

- 100% Take-home exam + presentation

Practicals

6 weeks 8hrs

- Theory
- Practicals
- Final assignment

Lesson Schedule

	Content			Tuesday 17-Feb	Intro to Data Engineering	Database management		week 3	Data Integration 24/02/2026 Jordy Vercammen	Data Engineering class 2	
Tuesday 10-Feb Steve Hillen	Introduction Dimensional Modelling Normal forms Exercise Flight Snowflake as a modelingtechnique			week 2 Jordy Vercammen	exercises on these						
	Multiple facttables Exercise Airport sales Slowly changing Dimensions Exercise SCD Extra exercise Callcenter Exercise Warehouse										
								week 6		mini project review bronze silver mini assignment	silver gold
Data integration 03-Mar Jordy Vercammen	put the things in practice	put it in practice model the silver layer		week 5 Data Vault 10-Mar Steve Hillen	Review Dimensional Model Data Vault Exercise Data Vault				17-Mar Jordy Vercammen		
	mini assignment bronze - silver	source to bronze to silver								mini project Exam Prep + explenation Exam	review
	mini project	modelling and bronze finished			review Data vault						

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Why Dimensional Modeling

What is Business intelligence

Business Intelligence (BI) refers to the technologies, strategies, and processes used to analyse business data and support decision-making.

It involves **data collection**, **visualization**, and **reporting** to uncover trends, patterns, and insights that drive strategic actions. BI helps organizations improve efficiency, optimize operations, and gain a competitive advantage.

Think of **Business Intelligence (BI) and Analytics** as the "translation layer" between raw, messy data and smart, winning decisions. In a world where every click, purchase, and sensor reading is recorded, these fields allow companies to stop guessing and start knowing.

Use of a BI environment

Use of a BI environment

- Management reporting
- Segmentation
- Customer overview
- Loyalty cards
- Fraud detection
- Dashboards and balanced scorecards
- Etc.

Why a Data warehouse

- “We have mountains of data in this company, but we can’t access it.”
- “We need to slice and dice the data every which way.”
- “You’ve got to make it easy for businesspeople to get at the data directly.”
- “Just show me what is important.”
- “It drives me crazy to have two people present the same business metrics at a meeting, but with different numbers.”
- “We want people to use information to support more fact-based decision making.”

Why a Data warehouse

- The data warehouse must make an organization's information **easily accessible**.
- The data warehouse must present the organization's information **consistently**.
- The data warehouse must be **adaptive** and **resilient to change**.
- The data warehouse must be a secure bastion that **protects our information assets**.
- The data warehouse must serve as the foundation for **improved decision making**.
- The **business community must accept** the data warehouse if it is to be deemed successful.

Why a Data warehouse

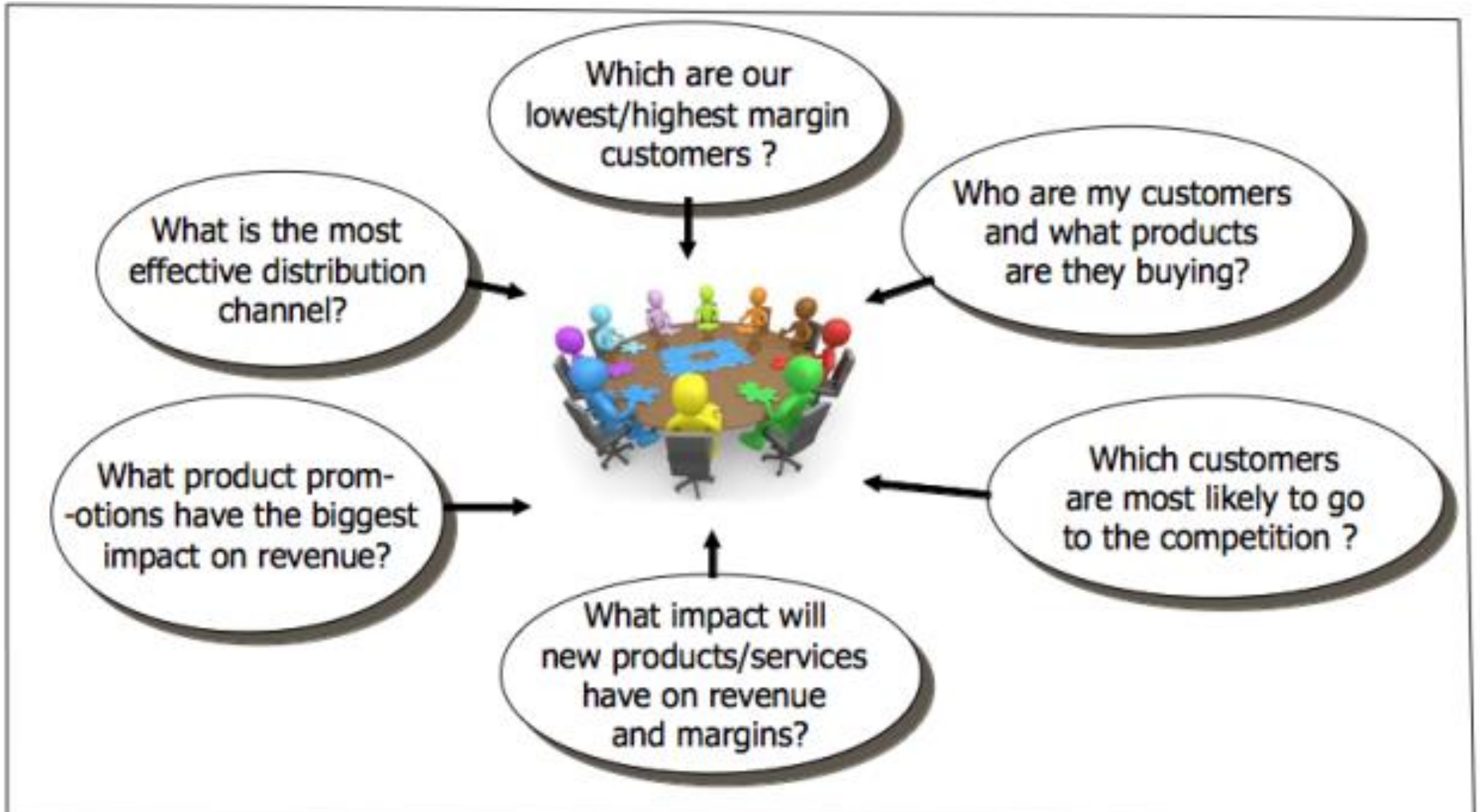
Data becomes information only if you

- Have the data and
- Know you have it and
- Can access it and
- Can trust it

Why do we use a data warehouse



To solve business questions



Why dimensional modeling

Operational databases are normalized

- Many related tables/entities
- No redundancy
- Complex queries for relatively simple questions
- Poor performance in complex queries

  ➔Not suitable for BI OLAP workload

Solution DWH: Dimensional modeling

- The technique used is denormalisation
- Less related tables/entities
- Redundant and aggregated data allowed
- Few joins are needed for complex queries, so performance increases

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MULTI-DIMENSIONAL MODELING

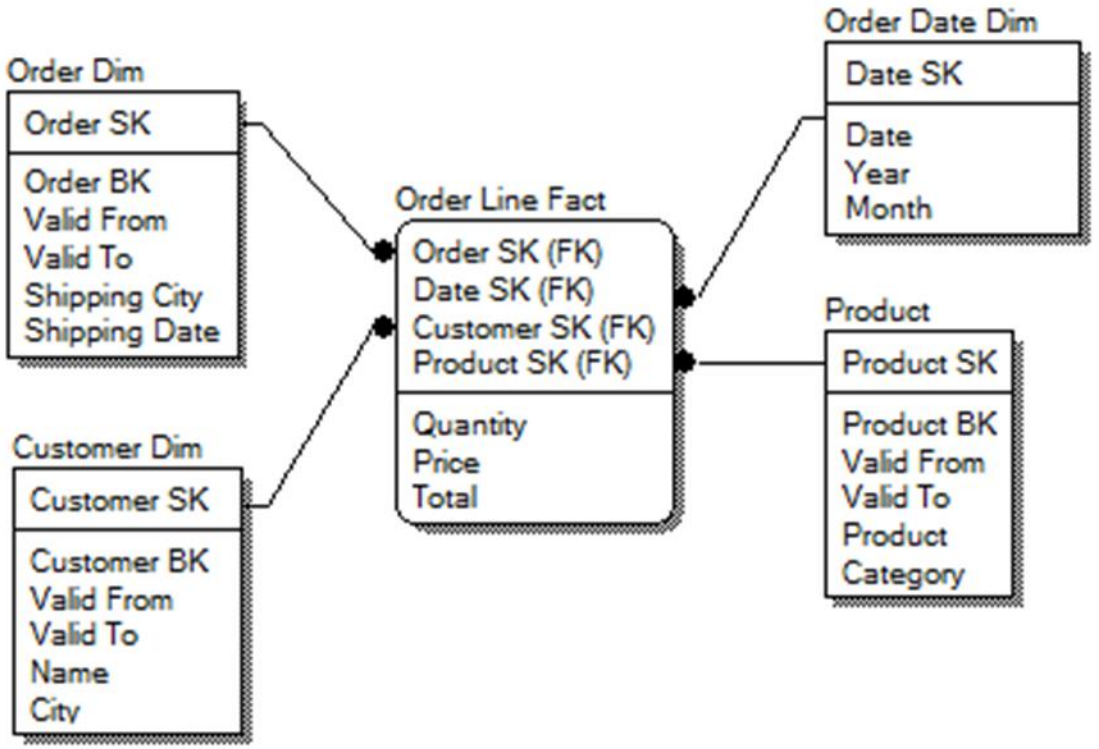
Typical techniques

Techniques

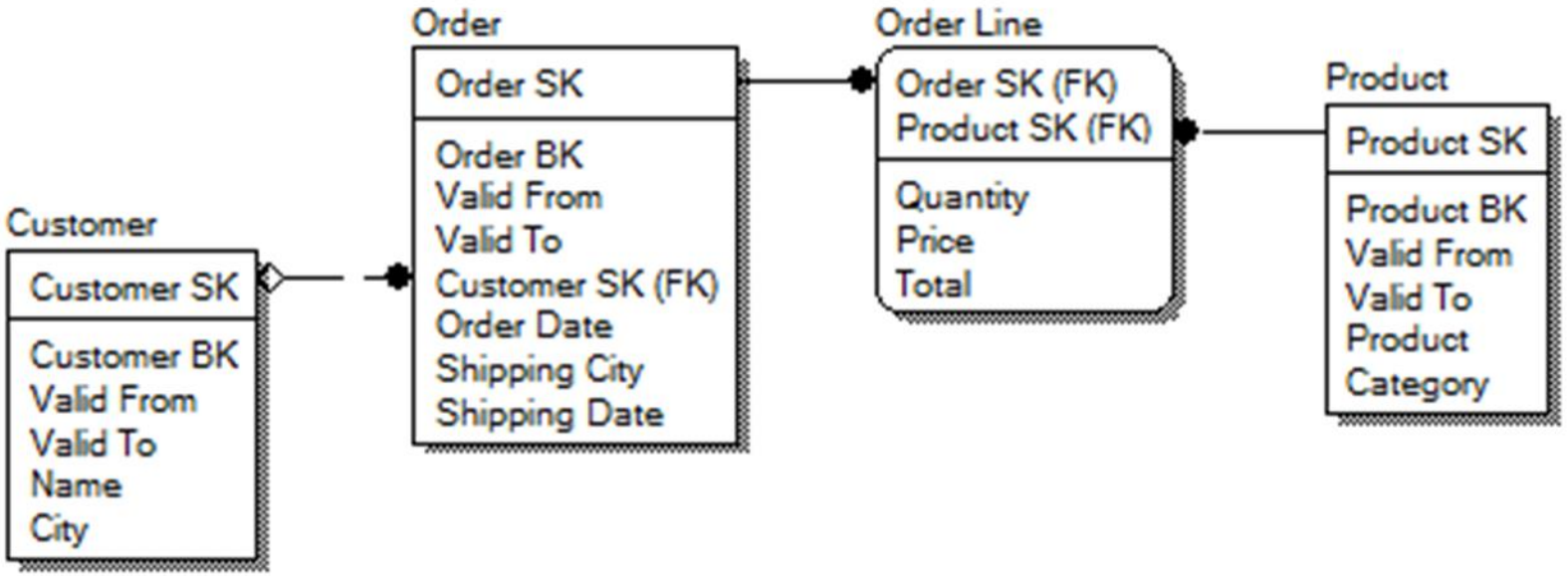
The most common modeling techniques:

- 3NF
  - Every table has a unique key, Non key attributes are dependent on the whole key. Non key attributes have to be dependent on nothing but the key.
- Dimensional modeling
  - Dimensions and facts.
- Data Vault
  - Data vault modeling is a database modeling method in which issues such as auditing, tracing of data, loading speed and resilience to change are emphasized. This means that every row in a data vault must be accompanied by record source and load date attributes, enabling an auditor to trace values back to the source.

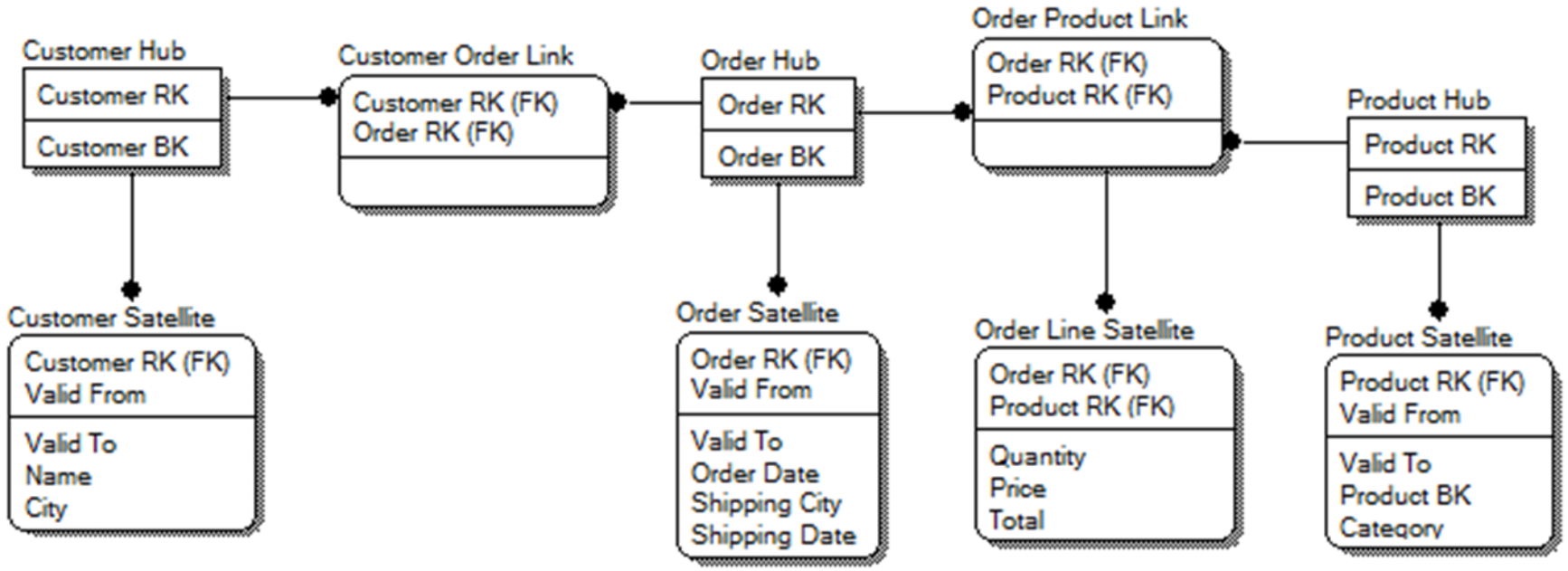
Techniques - Dimensional modeling



Techniques - 3NF

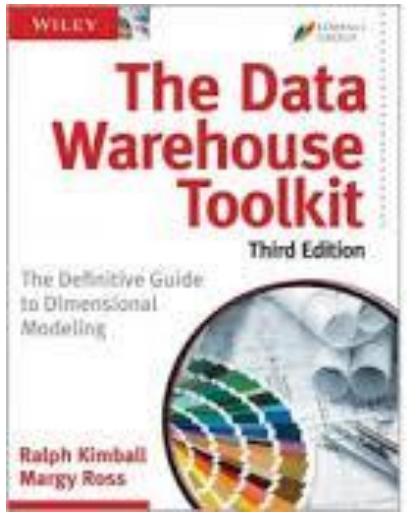


Techniques - Data Vault

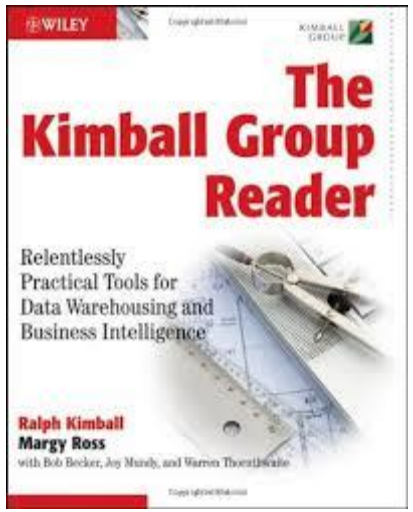


Why dimensional modeling

Founded by Ralph Kimball





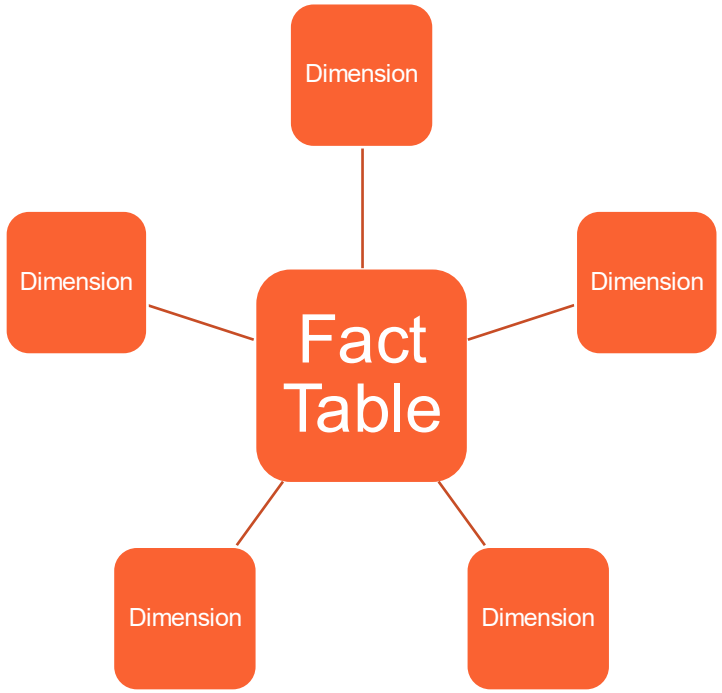


Kimballs modeling: starschema's and snowflakes

Dimensional datamodel: starschema

A starschema has 2 types of tables

- The fact(s)
- The dimension(s)

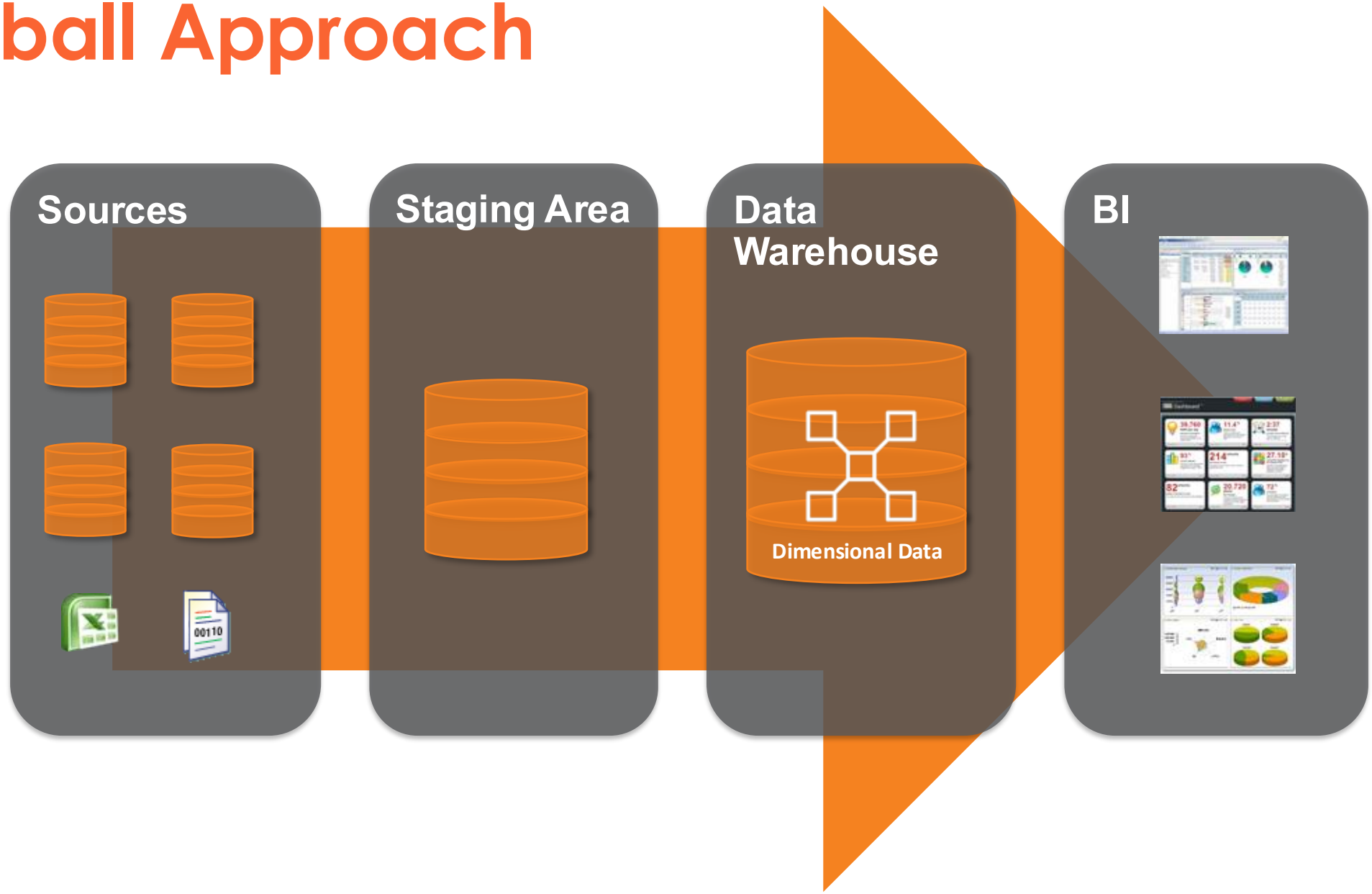


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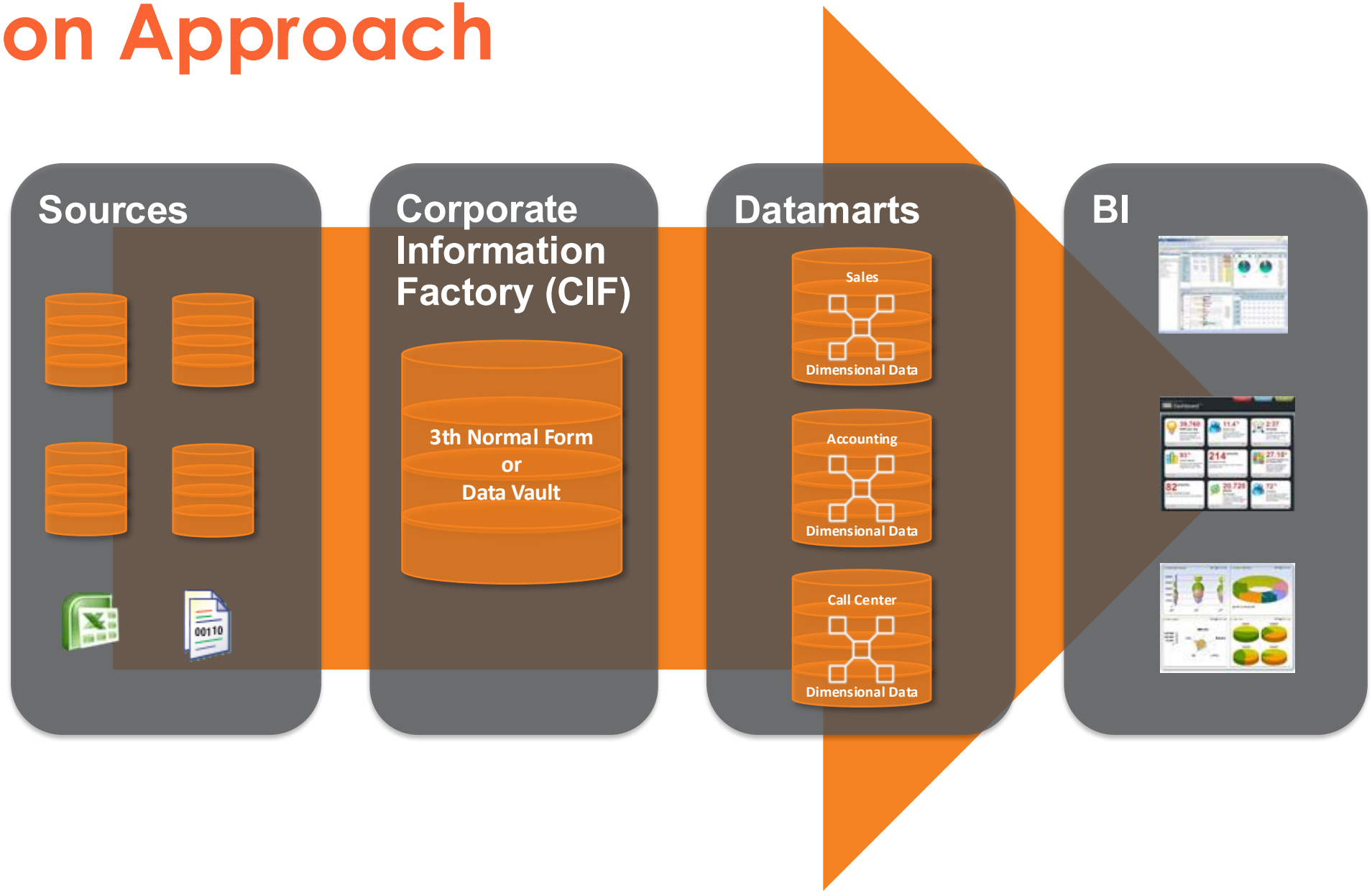
MULTI-DIMENSIONAL MODELING

Typical architectures

Data Warehouse architecture Kimball Approach



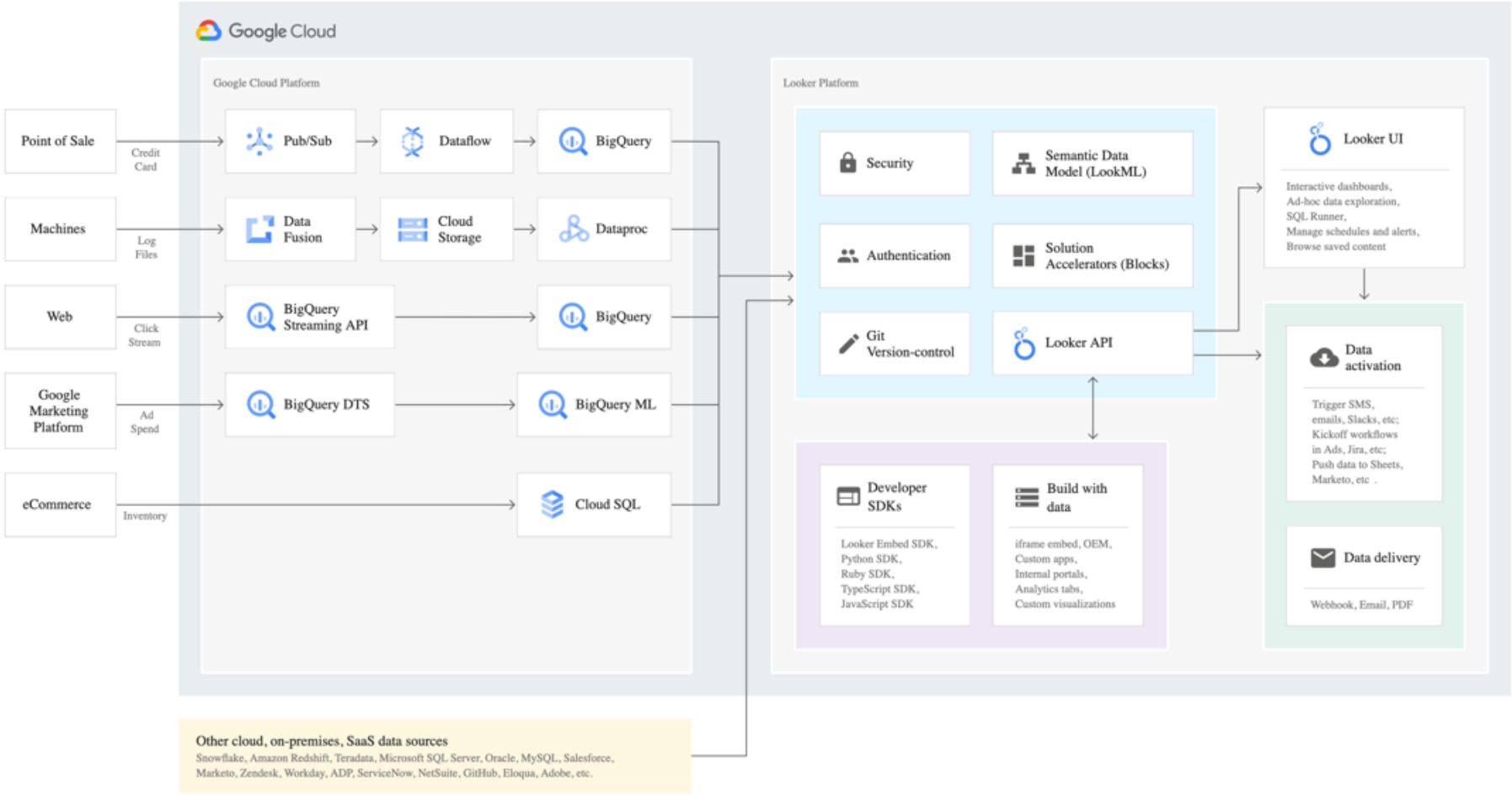
Data Warehouse architecture
Inmon Approach



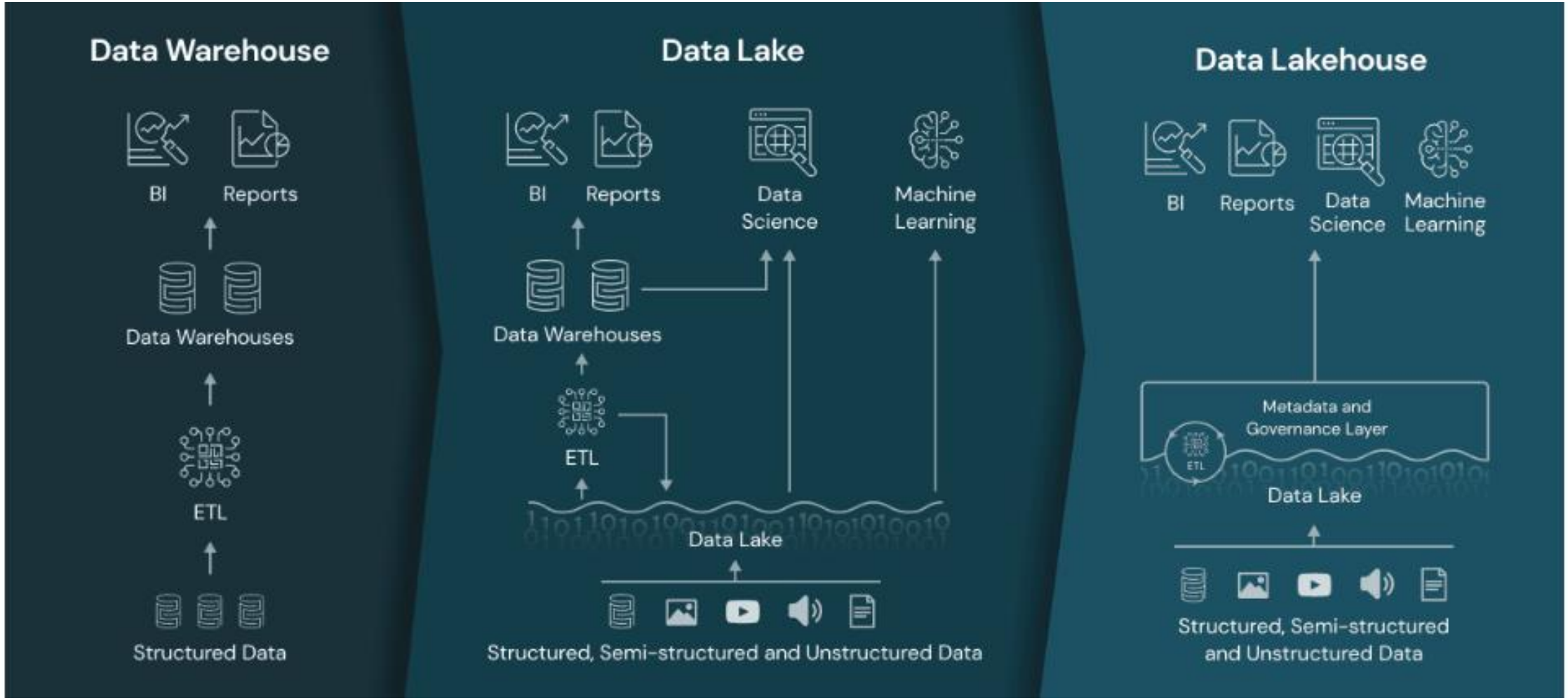
Analytics example by Looker



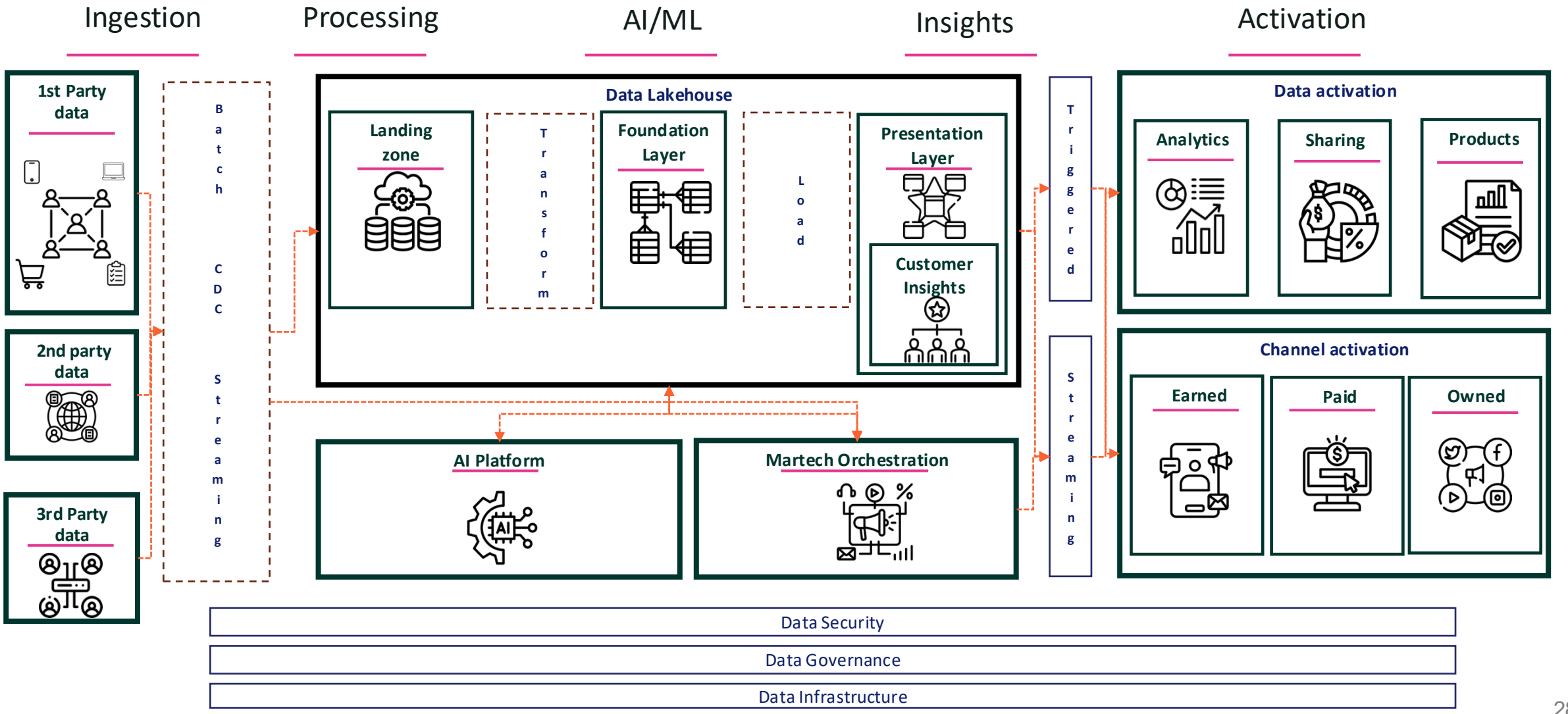
Many steps in dataprocessing



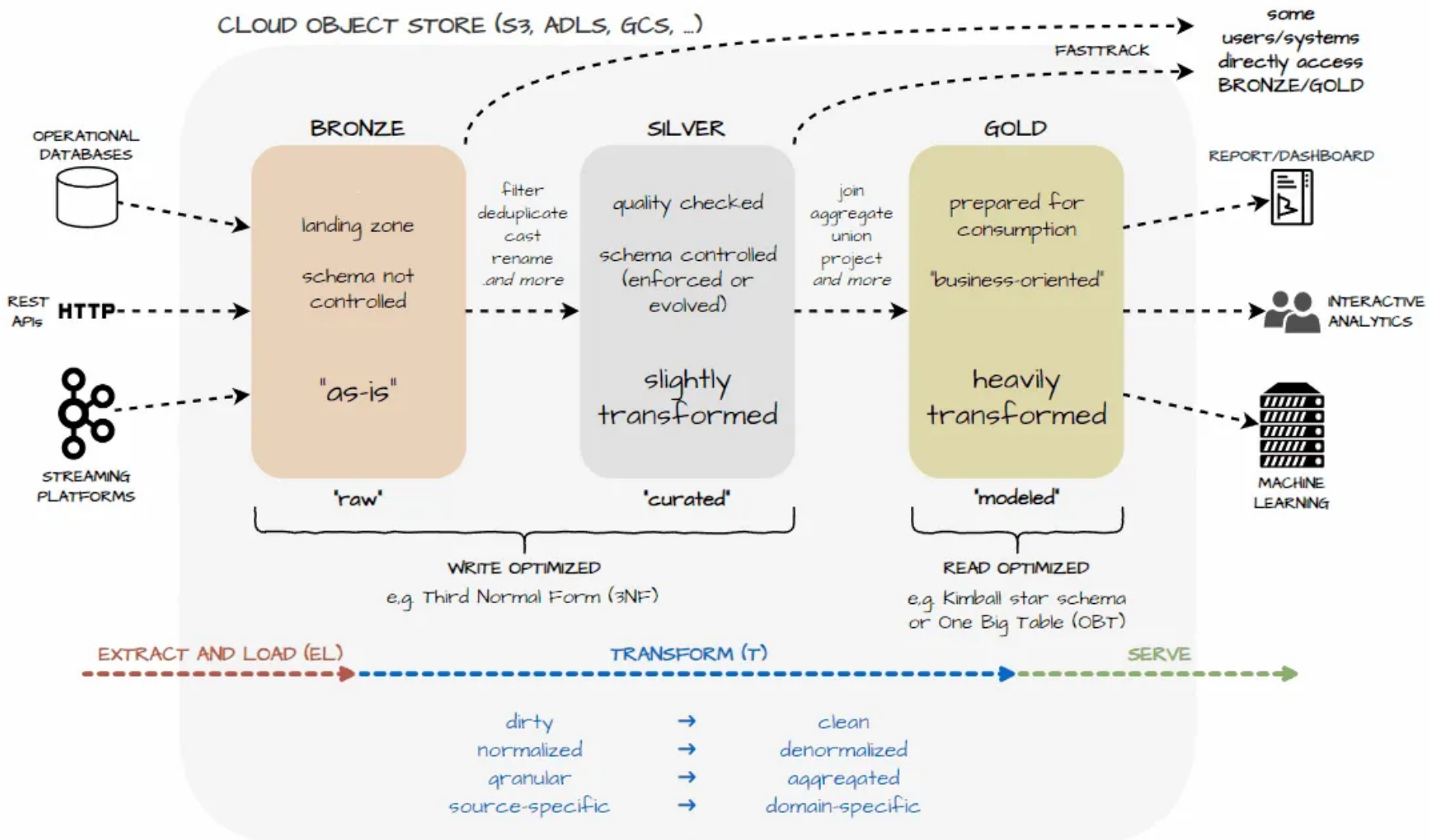
Different architectures



Data processing through a data lakehouse

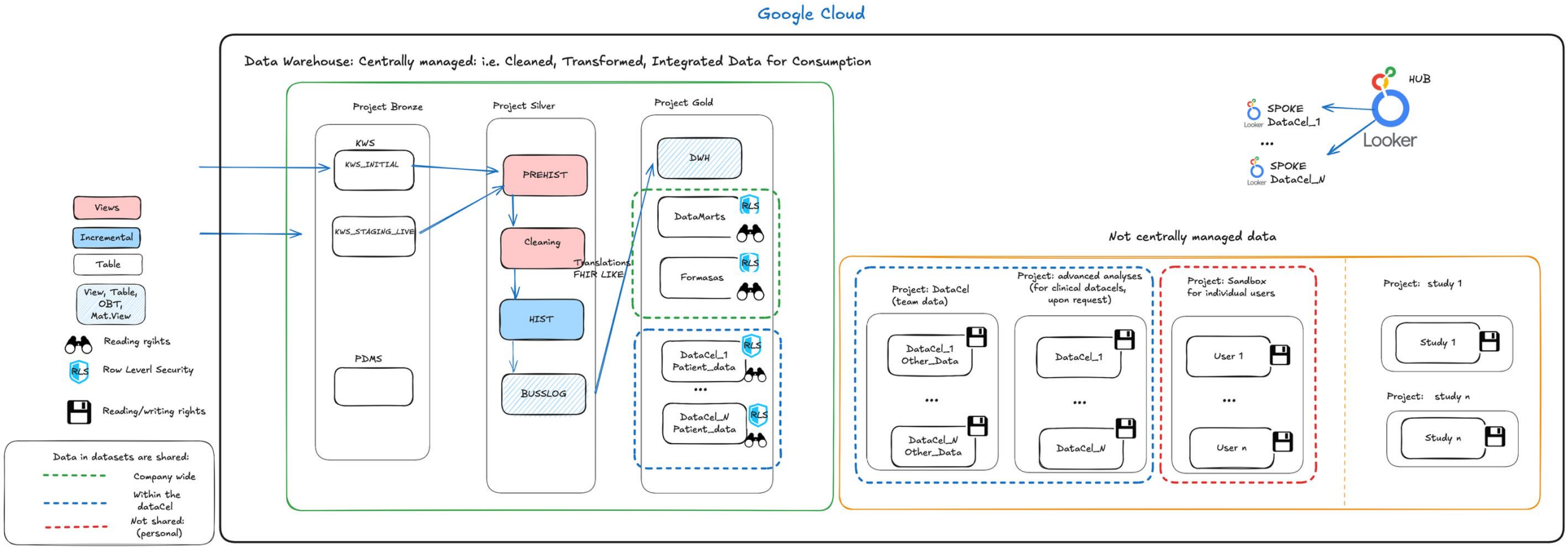


The Medallion Architecture

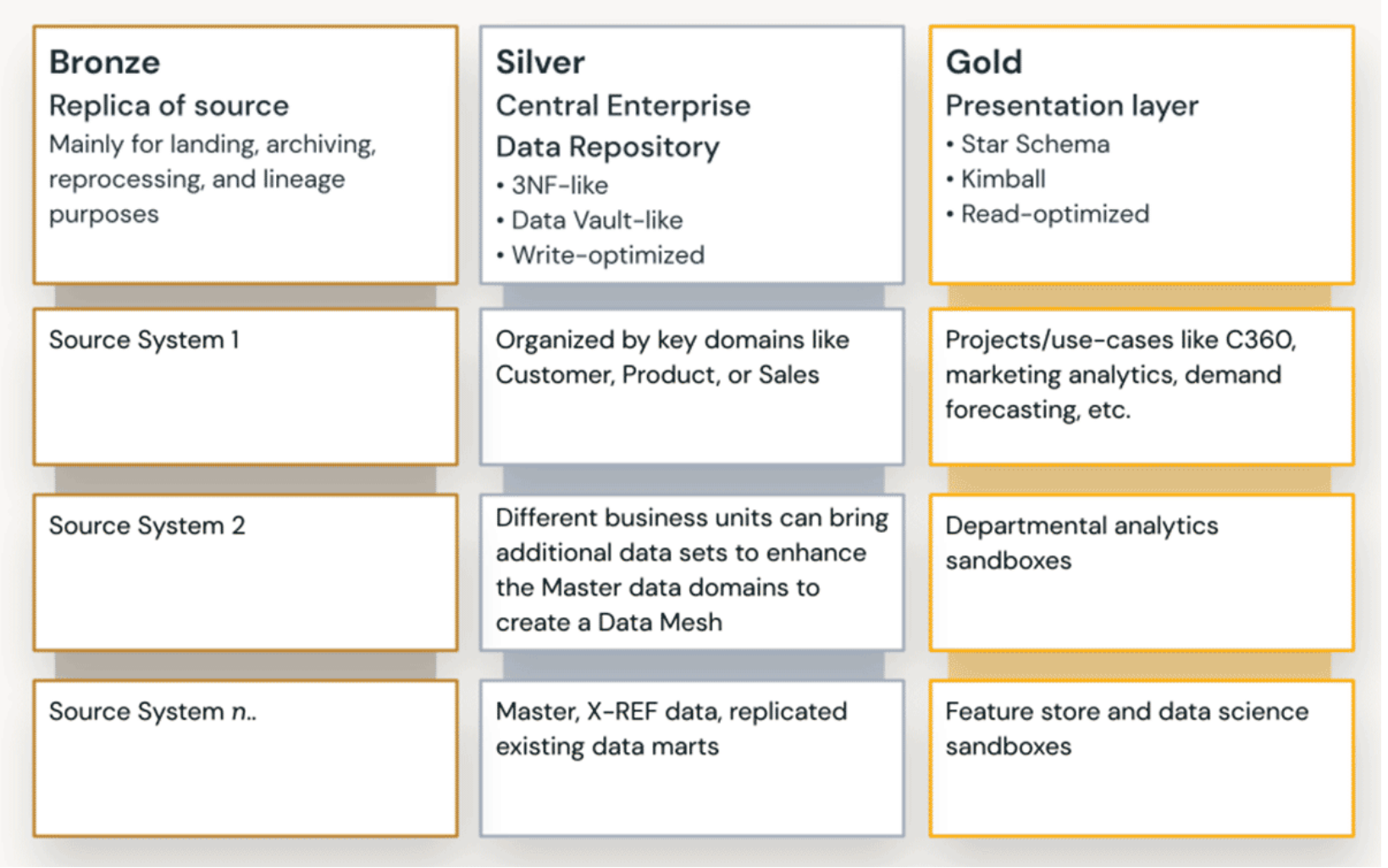


JORRIT SANDBRINK

GENERAL IDEA
MANY VARIATIONS EXIST


Google Cloud
Data Warehouse: Centrally managed: i.e. Cleaned, Transformed, Integrated Data for Consumption
Project Bronze
Project Silver
Project Gold
KWS
KWS_INITIAL
KWS_STAGING_LIVE
PDMS
PREHIST
Cleaning
HIST
BUSSLOG
Translations FHIR LIKE
DWH
DataMarts
Formasas
DataCel_1 Patient_data
DataCel_N Patient_data
RLS
HUB
SPOKE DataCel_1
SPOKE DataCel_N
Looker
Views
Incremental
Table
View, Table, OBT, Mat.View
Reading rgihts
Row Leverl Security
Reading/writing rights
Data in datasets are shared:
Company wide
Within the dataCel
Not shared: (personal)
Not centrally managed data
Project: DataCel (team data)
DataCel_1 Other_Data
DataCel_N Other_Data
Project: advanced analyses (for clinical datacels, upon request)
DataCel_1
DataCel_N
Project: Sandbox for individual users
User 1
User n
Project: study 1
Study 1
Project: study n
Study n

Data lakehouse architecture



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DIMENSIONAL MODELING

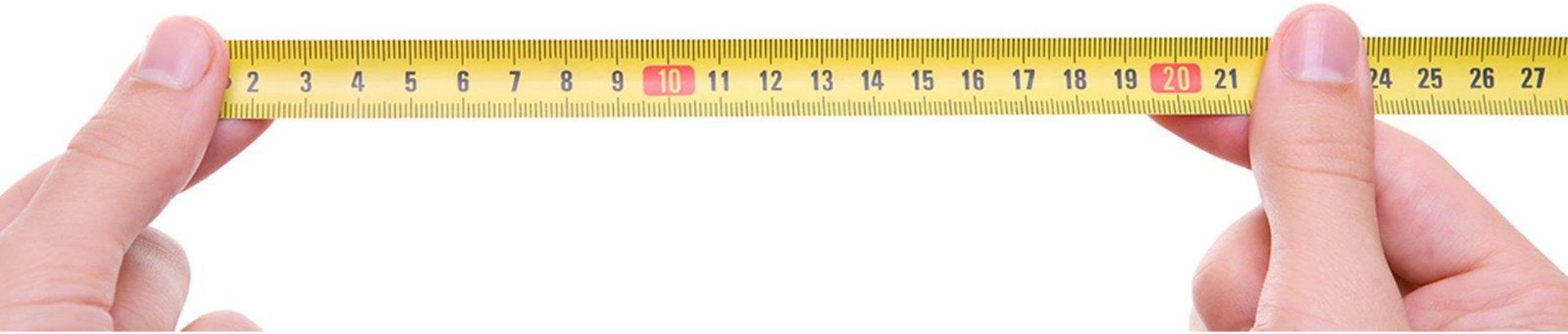
Base components of a Dimensional Model

Why do we use a Data Warehouse

Benefits of Data Driven Decision Making



Measures



5000

€5000

€5000
Salary


NO
CONTEXT

Provide dimensional context

By
For

BRING dimensional context

What is the monthly salary by deparment, by role
What is the average number of students subscribed by campus, by year and by major

BRING dimensional context

What is the monthly salary by deparment, by role
What is the average number of students subscribed by campus, by year and by major

Measures

BRING dimensional context

What is the monthly salary by deparment, by role
What is the average number of students subscribed by campus, by year and by major

Dimensional context

What about

What is the monthly salary for managers, by deparment, by role
What is the average number of for engineering students subscribed by campus, by year and by major

Dimensional context

Dimensional context “By” vs “For”

By: “Sliced and grouped” for the entire dimension
For: one or more specific value from within the entire dimension

Dimensional datamodel: starschema

The fact contains measures that make the business process measurable. KPI's are based on these numbers.

The numbers don't stand on itself. The context is provided via the dimensions.

Dimensional datamodel: starschema

Fact

- Number of students: 89

Meaning?

- Number of students subscribed for this lesson

Dimensional datamodel: starschema

Fact

- Number of students: 89

Meaning?

- Number of students subscribed for this lesson

Dimension: time
Dimension: Lesson

Dimensional datamodel: starschema

Fact

- Number of pieces sold: 72

Meaning?

- Number of iphone 15 sold on the February 17th at Apple store Brussels

Dimensional datamodel: starschema

Fact

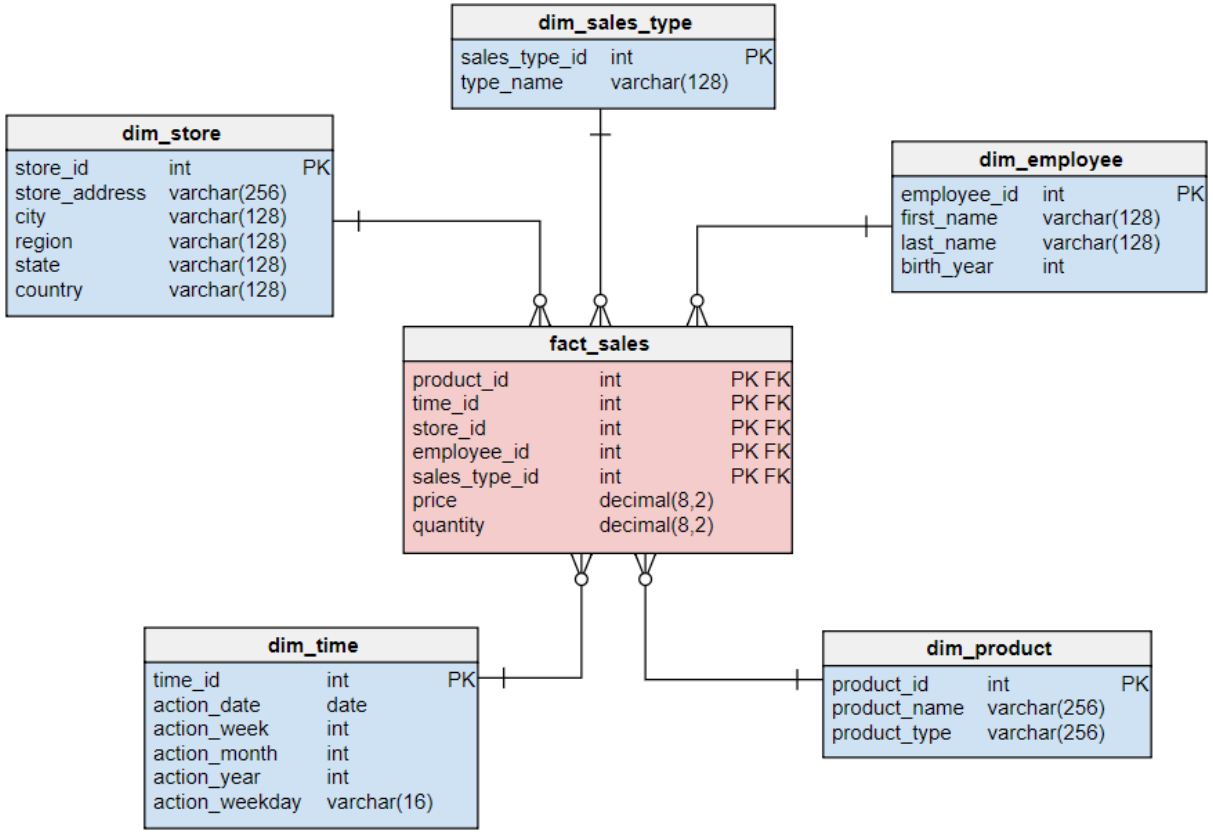
- Number of pieces sold: 72

Meaning?

- Number of iphone 15 sold on the February 17th at Apple store Brussels

Dimension: product
Dimension: time
Dimension: store

Example starschema



What is special at the dimension table?

Which answers can this schema provide?

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MULTI-DIMENSIONAL MODELING

Data normalization

Why Normalize the data

- Modeling the data warehouse
  - DWH Data model characteristics
    - Enterprise focus (versus department or function focus)
    - **Limit redundancy for simplifying the load**
    - Stable and independent of processes, technology, applications
    - Consistent to allow data integration and reconciliation
    - Flexible for data usage (reliable and solid foundation for any BI capability)
    - Designed to support quick and efficient load of data
    - Designed to support data quality processes (incl. error detection and correction), referential integrity checks, etc.
    - Designed to be independent of the technologies that will be “plugged” on the data warehouse tools, etc)

=> Data normalization seems to be the right technique to use...

What's data Redundancy

Student_id	Name	Surname	Course_id	CourseName
100	Jim	Black	C001	Coding
101	Amanda	Holland	C001	Coding
102	Marlin	Kilan	C001	Coding
103	Alice	Green	C001	Coding

What‘s data Redundancy

Student_id	Name	Surname	Course_id	CourseName
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100	Jim	Black	C001	Coding
101	Amanda	Holland	C001	Coding
102	Marlin	Kilan	C001	Coding
103	Alice	Green	C001	Coding

Insert anomaly

Student_id	Name	Surname	Course_id	CourseName
100	Jim	Black	C001	Coding
101	Amanda	Holland	C001	Coding
102	Marlin	Kilan	C001	Coding
103	Alice	Green	C001	Coding
104	Jason	Brown	C001	Coding

Delete anomaly

Update anomaly

What's data Redundancy

Student_id	Name	Surname	Course_id	CourseName
100	Jim	Black	C001	Coding
101	Amanda	Holland	C001	Coding
102	Marlin	Kilan	C001	Coding
103	Alice	Green	C001	Coding

Insert anomaly

Student_id	Name	Surname	Course_id	CourseName
100	Jim	Black	C001	Coding
101	Amanda	Holland	C001	Coding
102	Marlin	Kilan	C001	Coding
103	Alice	Green	C001	Coding
104	Jason	Brown	C001	Coding
105	Kim	Flower	✖	✖

Delete anomaly

Update anomaly

Can't insert rows with empty data

What's data Redundancy

Student_id	Name	Surname	Course_id	CourseName
100	Jim	Black	C001	Coding
101	Amanda	Holland	C001	Coding
102	Marlin	Kilan	C001	Coding
103	Alice	Green	C001	Coding

Insert anomaly

Student_id	Name	Surname	Course_id	CourseName
100	Jim	Black	C001	Coding
101	Amanda	Holland	C001	Coding
102	Marlin	Kilan	C001	Coding
103	Alice	Green	C001	Coding

Delete anomaly

Student_id	Name	Surname	Course_id	CourseName
100	Jim	Black	C001	Coding
101	Amanda	Holland	C001	Coding
~~102~~	~~Marlin~~	~~Kilan~~	C001	Coding
~~103~~	~~Alice~~	~~Green~~	C001	Coding

Update anomaly

What's data Redundancy

Student_id	Name	Surname	Course_id	CourseName
100	Jim	Black	C001	Coding
101	Amanda	Holland	C001	Coding
102	Marlin	Kilan	C001	Coding
103	Alice	Green	C001	Coding

Insert anomaly

Student_id	Name	Surname	Course_id	CourseName
100	Jim	Black	C001	Coding
101	Amanda	Holland	C001	Coding
102	Marlin	Kilan	C001	Coding
103	Alice	Green	C001	Coding

Delete anomaly

Student_id	Name	Surname	Course_id	CourseName
100	Jim	Black	C001	Coding
101	Amanda	Holland	C001	Coding
~~102~~	~~Marlin~~	~~Kilan~~	~~C001~~	~~Coding~~
~~103~~	~~Alice~~	~~Green~~	~~C001~~	~~Coding~~

Loose extra information

Update anomaly

What's data Redundancy

Student_id	Name	Surname	Course_id	CourseName
100	Jim	Black	C001	Coding
101	Amanda	Holland	C001	Coding
102	Marlin	Kilan	C001	Coding
103	Alice	Green	C001	Coding

Insert anomaly

Student_id	Name	Surname	Course_id	CourseName
100	Jim	Black	C001	Coding
101	Amanda	Holland	C001	Coding
102	Marlin	Kilan	C001	Coding
103	Alice	Green	C001	Coding

Delete anomaly

Student_id	Name	Surname	Course_id	CourseName
100	Jim	Black	C001	Coding
101	Amanda	Holland	C001	Coding
~~102~~	~~Marlin~~	~~Kilan~~	~~C001~~	~~Coding~~
~~103~~	~~Alice~~	~~Green~~	~~C001~~	~~Coding~~

Update anomaly

Student_id	Name	Surname	Course_id	CourseName
100	Jim	Black	C001	Java
101	Amanda	Holland	C001	Coding
102	Marlin	Kilan	C001	Java
103	Alice	Green	C001	Coding

Loose extra information

Solution

The solution: normal forms

- The most commonly used forms
  - First
  - Second
  - Third

Normalization: Pros and cons

Pros

- Reduce data redundancy & space required
- Enhance data consistency
- Enforce data integrity
- Reduce update cost
- Provide maximum flexibility in responding ad hoc queries
- Allow the use of parallelism,
- Can reduce the total number of rows per block.

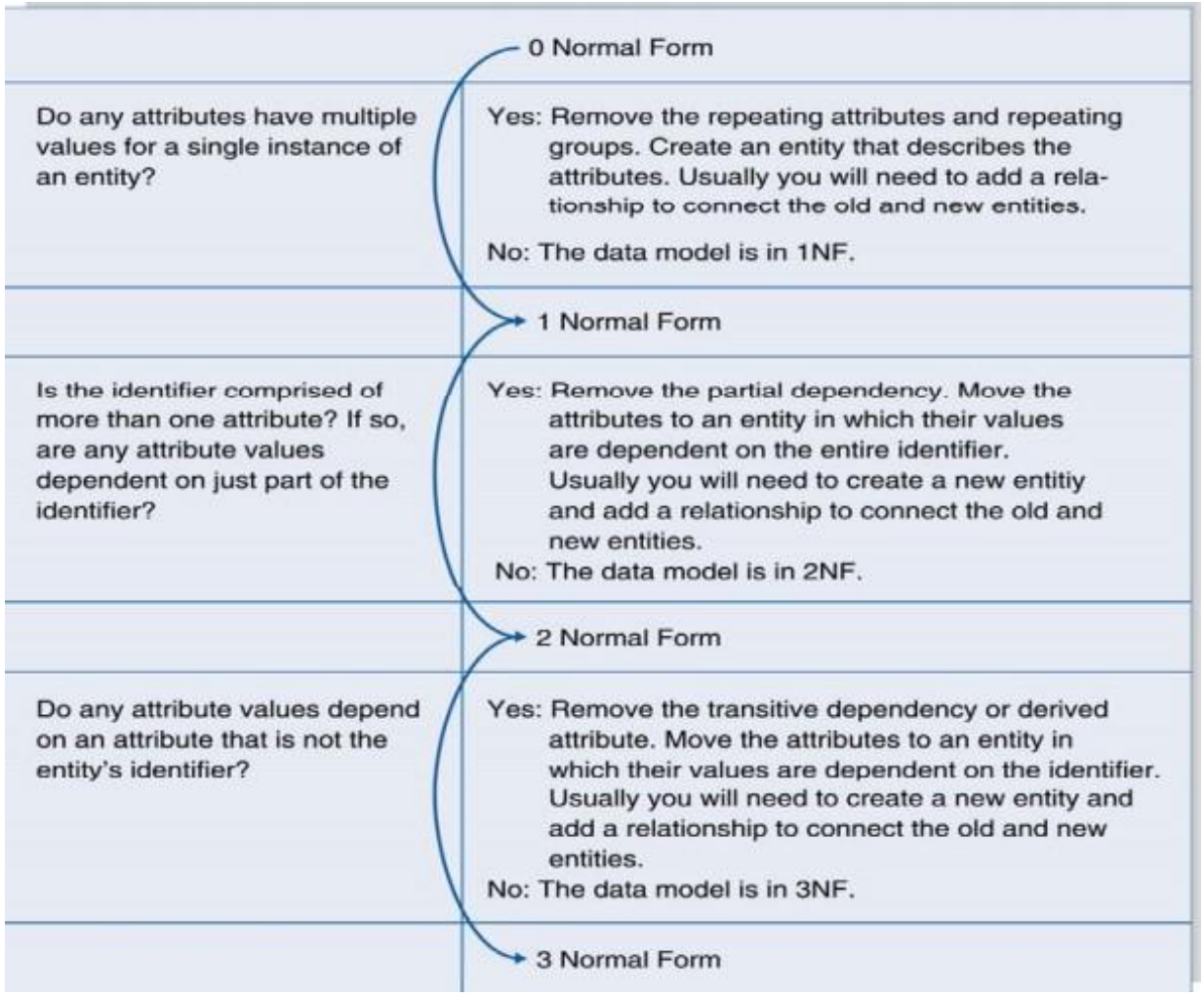
Improve:

- Software Design - Maintainability
- Reusability
- Readability

Cons

- Many complex queries will be slower because joins have to be performed to retrieve relevant data from several normalized tables
- Programmers/users have to understand the underlying data model of an database application in order to perform proper model of an database application in order to perform proper joins among several tables
- The formulation of multiple-level queries is a non-trivial task.

Normalization steps



Modeling the DWH – 0NF

Basetable
Contains al the required data

Project Code	Project Name	Project Manager	Project Budget	Employee No.	Employee Name	Department No.	Department Name	Hourly Rate
PC010	Reservation System	Mr. Ajay	120500	S100	Mohan	D03	Database	21.00
PC010	Reservation System	Mr. Ajay	120500	S101	Vipul	D02	Testing	16.50
PC010	Reservation System	Mr. Ajay	120500	S102	Riyaz	D01	IT	22.00
PC011	HR System	Mrs. Charu	500500	S103	Pavan	D03	Database	18.50
PC011	HR System	Mrs. Charu	500500	S104	Jitendra	D02	Testing	17.00
PC011	HR System	Mrs. Charu	500500	S315	Pooja	D01	IT	23.50
PC012	Attendance System	Mr. Rajesh	710700	S137	Rahul	D03	Database	21.50
PC012	Attendance System	Mr. Rajesh	710700	S218	Avneesh	D02	Testing	15.50
PC012	Attendance System	Mr. Rajesh	710700	S109	Vikas	D01	IT	20.50

UNF

Modeling the DWH – 1NF

A database table is said to be in 1NF if it contains no repeating fields/columns. The process of converting the UNF table into 1NF is as follows:

- Separate the repeating fields into new database tables along with the key from the unnormalized database table.
- The primary key of new database tables may be a composite key

Modeling the DWH – 1NF

Project Code	Project Name	Project Manager	Project Budget	Employee No.	Employee Name	Department No.	Department Name	Hourly Rate
PC010	Reservation System	Mr. Ajay	120500	S100	Mohan	D03	Database	21.00
PC010	Reservation System	Mr. Ajay	120500	S101	Vipul	D02	Testing	16.50
PC010	Reservation System	Mr. Ajay	120500	S102	Riyaz	D01	IT	22.00
PC011	HR System	Mrs. Charu	500500	S103	Pavan	D03	Database	18.50
PC011	HR System	Mrs. Charu	500500	S104	Jitendra	D02	Testing	17.00
PC011	HR System	Mrs. Charu	500500	S315	Pooja	D01	IT	23.50
PC012	Attendance System	Mr. Rajesh	710700	S137	Rahul	D03	Database	21.50
PC012	Attendance System	Mr. Rajesh	710700	S218	Avneesh	D02	Testing	15.50
PC012	Attendance System	Mr. Rajesh	710700	S109	Vikas	D01	IT	20.50

UNF

Separate the repeating fields into new database tables along with the key from the unnormalized database table.

Modeling the DWH – 1NF

Project Code	Project Name	Project Manager	Project Budget	Employee No.	Employee Name	Department No.	Department Name	Hourly Rate
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PC010	Reservation System	Mr. Ajay	120500	S101	Vipul	D02	Testing	16.50
PC010	Reservation System	Mr. Ajay	120500	S102	Riyaz	D01	IT	22.00
PC011	HR System	Mrs. Charu	500500	S103	Pavan	D03	Database	18.50
PC011	HR System	Mrs. Charu	500500	S104	Jitendra	D02	Testing	17.00
PC011	HR System	Mrs. Charu	500500	S315	Pooja	D01	IT	23.50
PC012	Attendance System	Mr. Rajesh	710700	S137	Rahul	D03	Database	21.50
PC012	Attendance System	Mr. Rajesh	710700	S218	Avneesh	D02	Testing	15.50
PC012	Attendance System	Mr. Rajesh	710700	S109	Vikas	D01	IT	20.50

UNF

Primary Key

Project Code	Project Name	Project Manager	Project Budget
PC010	Reservation System	Mr. Ajay	120500
PC011	HR System	Mrs. Charu	500500
PC012	Attendance System	Mr. Rajesh	710700

Composite Key (Unique Key)

Project Code	Employee No.	Employee Name	Department No.	Department Name	Hourly Rate
PC010	S100	Mohan	D03	Database	21.00
PC010	S101	Vipul	D02	Testing	16.50
PC010	S102	Riyaz	D01	IT	22.00
PC011	S103	Pavan	D03	Database	18.50
PC011	S104	Jitendra	D02	Testing	17.00
PC011	S315	Pooja	D01	IT	23.50
PC012	S137	Rahul	D03	Database	21.50
PC012	S218	Avneesh	D02	Testing	15.50
PC012	S109	Vikas	D01	IT	20.50

1NF

Separate the repeating fields into new database tables along with the key from the unnormalized database table.

Modeling the DWH – 2NF

A database table is said to be in 2NF if it is in 1NF and contains only those fields/columns that are functionally dependent(means the value of the field is determined by the value of another field(s)) on the primary key. In 2NF we remove the partial dependencies of any non-key field.

The process of converting the database table into 2NF is as follows:Remove the partial dependencies(A type of functional dependency where a field is only functionally dependent on the part of primary key) of any non-key field.

If field B depends on field A and vice versa. Also for a given value of B, we have only one possible value of A and vice versa, Then we put the field B into new database table where B will be the primary key and also marked as a foreign key in a parent table.

Modeling the DWH – 2NF

Primary Key

Project Code	Project Name	Project Manager	Project Budget
PC010	Reservation System	Mr. Ajay	120500
PC011	HR System	Mrs. Charu	500500
PC012	Attendance System	Mr. Rajesh	710700

Composite Key (Unique Key)

Project Code	Employee No.	Employee Name	Department No.	Department Name	Hourly Rate
PC010	S100	Mohan	D03	Database	21.00
PC010	S101	Vipul	D02	Testing	16.50
PC010	S102	Riyaz	D01	IT	22.00
PC011	S103	Pavan	D03	Database	18.50
PC011	S104	Jitendra	D02	Testing	17.00
PC011	S315	Pooja	D01	IT	23.50
PC012	S137	Rahul	D03	Database	21.50
PC012	S218	Avneesh	D02	Testing	15.50
PC012	S109	Vikas	D01	IT	20.50

1NF

The process of converting the database table into 2NF is as follows:Remove the partial dependencies(A type of functional dependency where a field is only functionally dependent on the part of primary key) of any non-key field.

Modeling the DWH – 2NF

Primary Key

Project Code	Project Name	Project Manager	Project Budget
PC010	Reservation System	Mr. Ajay	120500
PC011	HR System	Mrs. Charu	500500
PC012	Attendance System	Mr. Rajesh	710700

Composite Key (Unique Key)

Project Code	Employee No.	Employee Name	Department No.	Department Name	Hourly Rate
PC010	S100	Mohan	D03	Database	21.00
PC010	S101	Vipul	D02	Testing	16.50
PC010	S102	Riyaz	D01	IT	22.00
PC011	S103	Pavan	D03	Database	18.50
PC011	S104	Jitendra	D02	Testing	17.00
PC011	S315	Pooja	D01	IT	23.50
PC012	S137	Rahul	D03	Database	21.50
PC012	S218	Avneesh	D02	Testing	15.50
PC012	S109	Vikas	D01	IT	20.50

1NF

Primary Key

Project Code	Project Name	Project Manager	Project Budget
PC010	Reservation System	Mr. Ajay	120500
PC011	HR System	Mrs. Charu	500500
PC012	Attendance System	Mr. Rajesh	710700

Composite Key

Project Code	Employee No.	Hourly Rate
PC010	S100	21.00
PC010	S101	16.50
PC010	S102	22.00
PC011	S103	18.50
PC011	S104	17.00
PC011	S315	23.50
PC012	S137	21.50
PC012	S218	15.50
PC012	S109	20.50

Primary Key

Employee No.	Employee Name	Department No.	Department Name
S100	Mohan	D03	Database
S101	Vipul	D02	Testing
S102	Riyaz	D01	IT
S103	Pavan	D03	Database
S104	Jitendra	D02	Testing
S315	Pooja	D01	IT
S137	Rahul	D03	Database
S218	Avneesh	D02	Testing
S109	Vikas	D01	IT

2NF

Modeling the DWH – 3NF

A database table is said to be in 3NF if it is in 2NF and all non-keys fields should be dependent on primary key or We can also say a table to be in 3NF if it is in 2NF and no fields of the table are transitively functionally dependent on the primary key. The process of converting the table into 3NF is as follows:

- Remove the transitive dependencies(A type of functional dependency where a field is functionally dependent on the Field that is not the primary key. Hence its value is determined, indirectly by the primary key)

- Make a separate table for transitive dependent Field.

Modeling the DWH – 3NF

Primary Key

Project Code	Project Name	Project Manager	Project Budget
PC010	Reservation System	Mr. Ajay	120500
PC011	HR System	Mrs. Charu	500500
PC012	Attendance System	Mr. Rajesh	710700

Composite Key

Project Code	Employee No.	Hourly Rate
PC010	S100	21.00
PC010	S101	16.50
PC010	S102	22.00
PC011	S103	18.50
PC011	S104	17.00
PC011	S315	23.50
PC012	S137	21.50
PC012	S218	15.50
PC012	S109	20.50

Primary Key

Employee No.	Employee Name	Department No.	Department Name
S100	Mohan	D03	Database
S101	Vipul	D02	Testing
S102	Riyaz	D01	IT
S103	Pavan	D03	Database
S104	Jitendra	D02	Testing
S315	Pooja	D01	IT
S137	Rahul	D03	Database
S218	Avneesh	D02	Testing
S109	Vikas	D01	IT

2NF

Remove the transitive dependencies

Modeling the DWH – 3NF

Primary Key

Project Code	Project Name	Project Manager	Project Budget
PC010	Reservation System	Mr. Ajay	120500
PC011	HR System	Mrs. Charu	500500
PC012	Attendance System	Mr. Rajesh	710700

Composite Key

Project Code	Employee No.	Hourly Rate
PC010	S100	21.00
PC010	S101	16.50
PC010	S102	22.00
PC011	S103	18.50
PC011	S104	17.00
PC011	S315	23.50
PC012	S137	21.50
PC012	S218	15.50
PC012	S109	20.50

Primary Key

Employee No.	Employee Name	Department No.	Department Name
S100	Mohan	D03	Database
S101	Vipul	D02	Testing
S102	Riyaz	D01	IT
S103	Pavan	D03	Database
S104	Jitendra	D02	Testing
S315	Pooja	D01	IT
S137	Rahul	D03	Database
S218	Avneesh	D02	Testing
S109	Vikas	D01	IT

2NF

Primary Key

Project Code	Project Name	Project Manager	Project Budget
PC010	Reservation System	Mr. Ajay	120500
PC011	HR System	Mrs. Charu	500500
PC012	Attendance System	Mr. Rajesh	710700

Composite Key

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PC010	S101	16.50
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PC011	S103	18.50
PC011	S104	17.00
PC011	S315	23.50
PC012	S137	21.50
PC012	S218	15.50
PC012	S109	20.50

Primary Key

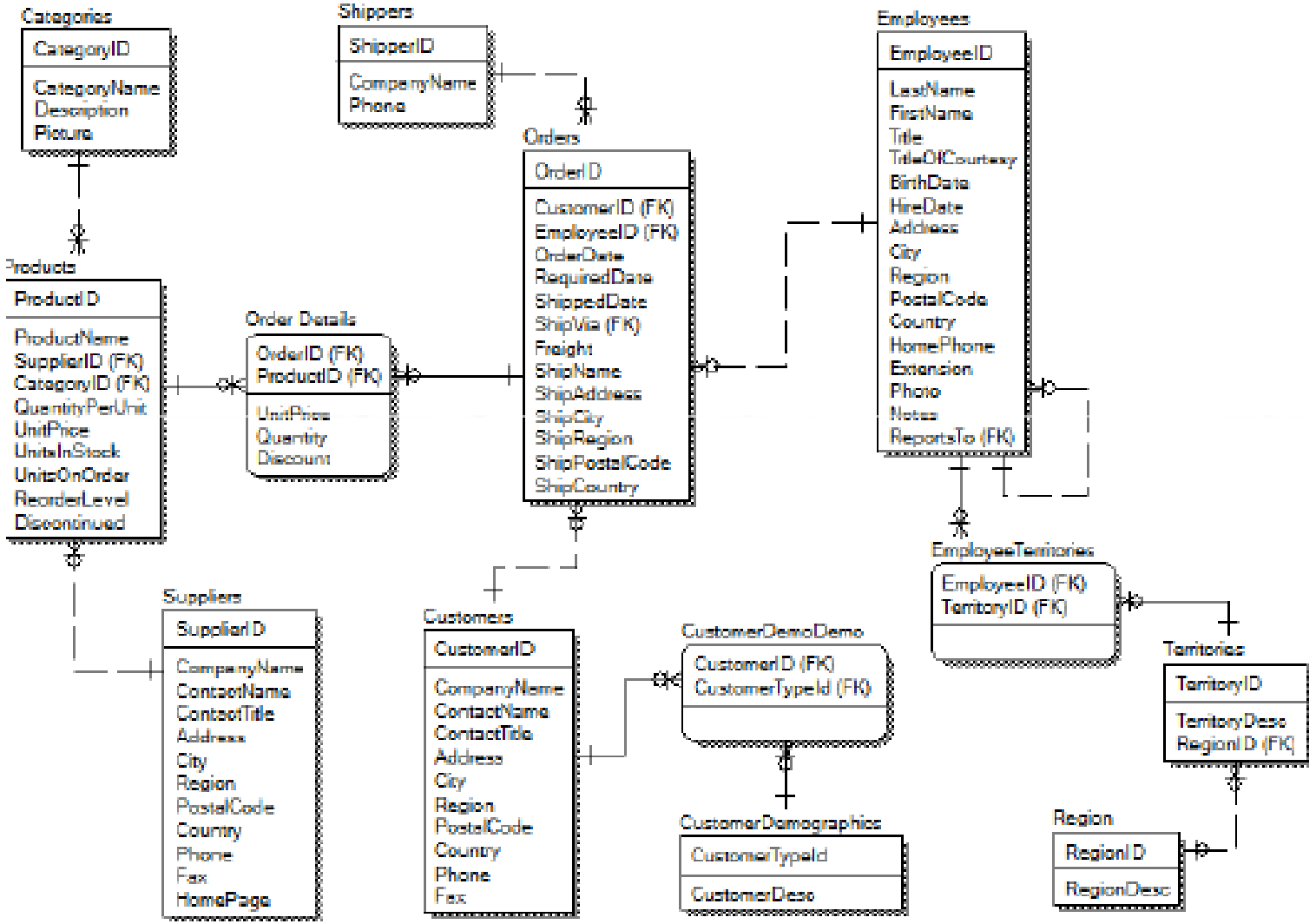
Employee No.	Employee Name	Department No.
S100	Mohan	D03
S101	Vipul	D02
S102	Riyaz	D01
S103	Pavan	D03
S104	Jitendra	D02
S315	Pooja	D01
S137	Rahul	D03
S218	Avneesh	D02
S109	Vikas	D01

Primary Key FK_Relationship

Department No.	Department Name
D01	IT
D02	Testing
D03	Database

3NF

Example of a normalized model (source system)



Normalization Exercise

Student_id	Name	Surname	State	Country	CourseName
100	Jim	Black	LA	USA	Coding, Accounting
101	Amanda	Holland	NY	USA	Coding
102	Marlin	Kilan	Berlin	Germany	Coding, Java
103	Alice	Green	London	UK	Coding

Normalization Exercise

Student_id	Name	Surname	State	Country	CourseName
100	Jim	Black	LA	USA	Coding, Accounting
101	Amanda	Holland	NY	USA	Coding
102	Marlin	Kilan	Berlin	Germany	Coding, Java
103	Alice	Green	London	UK	Coding

Student_id	Name	Surname	State_id
100	Jim	Black	1
101	Amanda	Holland	2
102	Marlin	Kilan	3
103	Alice	Green	4

State_id	State	Country
1	LA	USA
2	NY	USA
3	Berlin	Germany
4	London	UK

Student_id	Course_id
100	C001
100	C002
101	C001
102	C001
102	C003
103	C001

course_id	course
C001	Coding
C002	Accounting
C003	Java

Summary

1NF

- No repeating data/groups
- Each table must have a primary key

2NF

- Attributes have to be dependent on the whole key

3 NF

- Attributes have to be dependent on nothing but the key

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MULTI-DIMENSIONAL MODELING

Exercise Normal forms

ACME

Acme, Inc.
103 Majestic Way
Raleigh, NC 27606

Invoice Number #1785439

Issue Date 2018-02-01

Due Date 2018-02-01

DUE **$210.07**

Bill To:
Nathanael Erdman
75839 Nolan Crossroad
North Lethamouth, TX 78205

Item	Quantity	Unit Price	Amount
Pro[1] 02/01/2018 - 03/01/2018	1	$99.00	$99.00
IP Addresses	10	$1.00	$10.00
Minutes[2]	1398	$0.10	$139.80

Discount(s) Applied:

1. 25% Off Plan Fee	$24.75
2. 10% Off Minutes	$13.98

Subtotal	$248.80
Discounts	($38.73)
Total	$210.07
Paid	$0.00
Due	$210.07

0NF

Invoice (Name, address, city, postalcode, Invoicenumber, linenumber, issue date, due date, customer, customer address, Item, quantity, unit price, amount, discount)

1NF – remove repeating groups

Invoice (Name, address, city, postalcode, Invoicenumber, issue date, due date, customer, customer address)

Invoice detail(Invoicenumber*, linenumber, Item, quantity, unit price, amount, discount)

2NF – directly related to primary key

Invoice (Name, address, city, postalcode, Invoicenumber, issue date, due date, customer*)

Customer (Customer, customer address)

Invoice detail(Invoicenumber*,linenumber, Item, quantity, unit price, amount, discount)

3NF – Remove the transitive dependencies

Invoice (Name*, Invoicenumber, issue date, due date, customer*)

Company (Name, address, city, postalcode)

Customer (Customer, customer address)

Invoice detail(Invoicenumber*, linenumber, Item*, quantity, amount, discount)

Item (Item, unit price)

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Normalization vs Denormalization

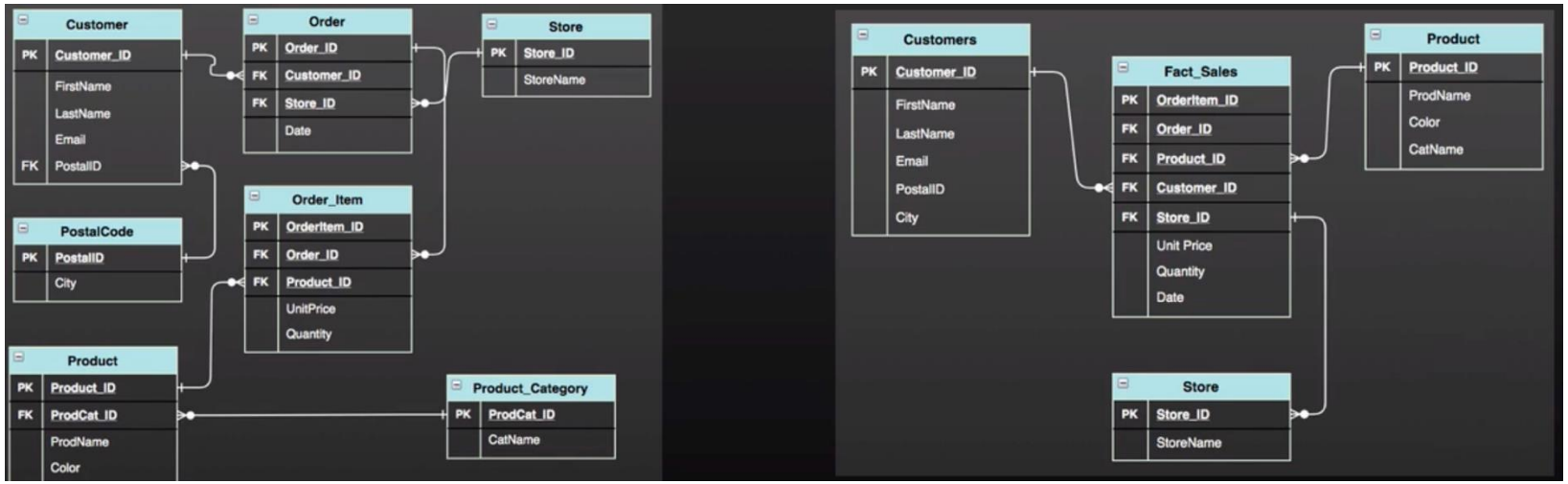
Normalization vs denormalization

Normalization

Divides larger tables into smaller tables to reduce redundancy
Increase number of tables and joins
Speed of update/delete is higher
OLTP system use for faster write

Denormalization

Combined tables into larger tables
Less tables means less joins
Speed of read is important
OLAP



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MULTI-DIMENSIONAL MODELING

Steps towards a starschema

4 steps towards a starschema

1. Choose the business process

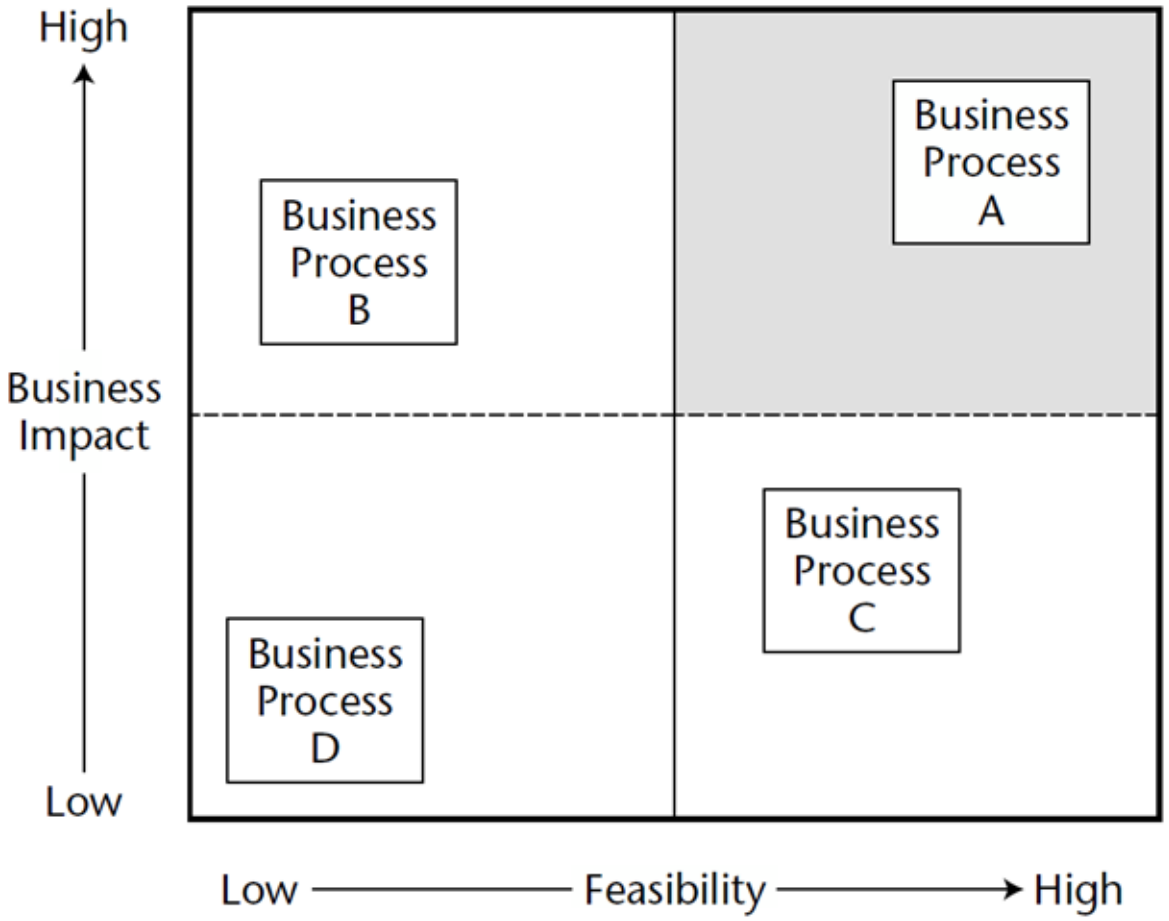
2. Declare the grain
   - What does the fact record mean?

3. Identify the dimensions
   - All descriptive context that has a single value in the presence of the measurement

4. Identify the facts and measures
   - Numeric additive measurements

5. Determine and define dimensional hierarchies and attributes

Prioritization



1. Define the Process

Normalizing the data => find relationships between the data
Dimensional modeling is looking at the business process

- Sales
- Marketing
- Follow-up payments
- Recruitmentprocedure
- …

2. Declaire the Grain

Kimball uses the grain statement
The grain is the most detailed level of a datamart

A more detailled analysis than the grain is not possible
Example: date/time

- Analyse data by year – by month – by week – by day – by hour – by minute - …

2. Declare the Grain

The grain can be derived directly from the information needs of the business
Example of the new students of Thomas More analysis

- 9 students from De naeyer institute
- 9 students from De naeyer institute where 3 students from Accountancy and 6 from IT

Example sales analysis

- Marketing whant's to create more campaings to sponsor more events or distribute more promotional flyers
- Sales wants a detailled overview per customer

2. Declare the Grain

The more detailled

- The more posibilities to analyse the data
- The more problems with performance
- The more storage is required

2. Declare the Grain

Assume that a store has 100 orders per day and per order 2 products are sold
Grain day

- monthly : 100*2*30 = 6 000 records
- yearly: 6 000 * 12 = 72 000 records
- After 10 year: 720 000 records

Grain Month

- Monthly: 1 record
- yearly: 12 records
- After 10 jaar: 120 records

3. Define Dimensions

Most important component within starschema
Dimensions gives a meaning to the numbers in the fact
The dimensions will be used to lookup data in the fact
The dimensions are used to limit (where) and group (group by) data in a query on the fact

3. Define dimensions

“Show monthly (GROUP BY) and overview of all sold Iphone X (WHERE) colourcode Black (WHERE)”

Dimension: Time
Dimension : Product
Dimension : Colour → Colour will be an attribute of the productdimension

3. Define Dimensions

Attributes define the quality of the dimension

The more attributes, the more analyses possible

Attributes contains ideally textual information and not numbers or codes

3. Define Dimensions

The key of a dimension is a surrogate key this means an autonumber without any meaning

To keep the link with the sourcedata a common technique is to include a alternate key of source key

- CustomerSourceKey

3. Define Dimensions

A Date dimension is available in almost all schema's
Why isn't there a date in the fact table?

- Filterfunction
  - For every relevant period (year, month, quarter, weeknumber, daynumber) an attribute
  - Do not include in every query DatePart (Quarter, [date]) by directly as a quarter in the dimension
- calenderyear is not a bookingyear
  - Define bookingyear is companyspecific (businessrule)
  - Foresee Attribute Bookingyear in the dimension

3. Define Dimensions

Why don’t we include a date in the fact?

- You can include extra attributes like holidays, closingperiod, …
  - For example to analyse the revenu at a public holiday
- Datedimensions contains records for every day (depending of the grain) in a year
  - Analyserapports will always use full month/year data without blanco’s
  - Example:no sales during the closing of a construction firm in the holiday period → july isn’t shown in the reports as there are no figures

4. Define facts

Facts are also called as measures as facts are

- numeric
- measurable
- comparable

A facttable is a collection of Foreign Keys to the dimensionrecords and the facts itself.
The facts within the facttable are all numeric measures that need to be calculated. But there are some exceptions which will be discussed later in the course

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MULTI-DIMENSIONAL MODELING

Sales example

1. Select the business process



HOTEL UNIVERSAL RESORT
RJ SHIPS AND GIFT STORE

11/11/15 10:21am Ticket No: 110595

T:01 Shft:1 Cshr:829 Guests:
Salesprsn:SMITH

ICE		3.00
COOLER 4 GAL		8.95
ARIZONA TEA	2.00 @ 2.75	5.50
NECKACE SILVER/ BLUE		25.00
10% DISCOUNT		(2.50)
PURSES (18.00)		18.00
Sub-total...:		60.45
Discount		2.50-
TAX		4.35
Total:		62.30

PAYMENT- CASH

THANK YOU

Notice: There no refunds on sale items.
Receipt/proof of purchase is required
for exchange.

1. Select the business process



HOTEL UNIVERSAL RESORT
RJ SHIPS AND GIFT STORE

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Total:		62.30

PAYMENT- CASH

THANK YOU

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for exchange.

- Sales or Tickets

2. Declare the grain

```
      HOTEL UNIVERSAL RESORT
       RJ SHIPS AND GIFT STORE


11/11/15 10:21am     Ticket No: 110595
----------------------------------------
T:01 Shft:1 Cshr:829         Guests:
Salesprsn:SMITH
----------------------------------------
ICE                                 3.00
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NECKACE SILVER/ BLUE               25.00
            10% DISCOUNT           (2.50)
PURSES (18.00)                     18.00
                                 ----------
                Sub-total...:      60.45
                Discount            2.50-
                TAX                 4.35
                                 ----------

                      Total:       62.30


PAYMENT- CASH


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```

2. Declare the grain



```
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       RJ SHIPS AND GIFT STORE


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ARIZONA TEA       2.00 @  2.75      5.50
NECKACE SILVER/ BLUE               25.00
            10% DISCOUNT           (2.50)
PURSES (18.00)                     18.00
                                 ----------
                Sub-total...:      60.45
                Discount            2.50-
                TAX                 4.35
                                 ----------

                      Total:       62.30


PAYMENT- CASH


THANK YOU


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```

- Cash receipt (line)

3. Choose the dimensions



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RJ SHIPS AND GIFT STORE

11/11/15 10:21am Ticket No: 110595

T:01 Shft:1 Cshr:829 Guests:
Salesprsn:SMITH

ICE		3.00
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Sub-total...:		60.45
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TAX		4.35
Total:		62.30

PAYMENT- CASH

THANK YOU

Notice: There no refunds on sale items.
Receipt/proof of purchase is required
for exchange.

3. Choose the dimensions

- Store
- Product
- Customer
- Employee
- Currency
- Time
- Discount



```
      HOTEL UNIVERSAL RESORT
       RJ SHIPS AND GIFT STORE


11/11/15 10:21am     Ticket No: 110595
----------------------------------------
T:01 Shft:1 Cshr:829         Guests:
Salesprsn:SMITH
----------------------------------------
ICE                                3.00
COOLER 4 GAL                       8.95
ARIZONA TEA       2.00 @  2.75      5.50
NECKACE SILVER/ BLUE              25.00
          10% DISCOUNT            (2.50)
PURSES (18.00)                    18.00
                                 ----------
                Sub-total...:     60.45
                Discount           2.50-
                TAX                4.35
                                 ----------

                      Total:      62.30


PAYMENT- CASH


THANK YOU


Notice: There no refunds on sale items.
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```

4. Identify the facts

```
      HOTEL UNIVERSAL RESORT
       RJ SHIPS AND GIFT STORE


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                                  ------
                Sub-total...:      60.45
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                                  ------

                      Total:       62.30


PAYMENT- CASH


THANK YOU


Notice: There no refunds on sale items.
Receipt/proof of purchase is required
for exchange.
```

4. Identify the facts

Fact = Sales

Measures:

- Quantity Sold
- Sales Amount

- Discount?
- Tax?



```
      HOTEL UNIVERSAL RESORT
       RJ SHIPS AND GIFT STORE


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                                 ----------
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                Discount           2.50-
                TAX                4.35
                                 ----------

                      Total:      62.30


PAYMENT- CASH


THANK YOU

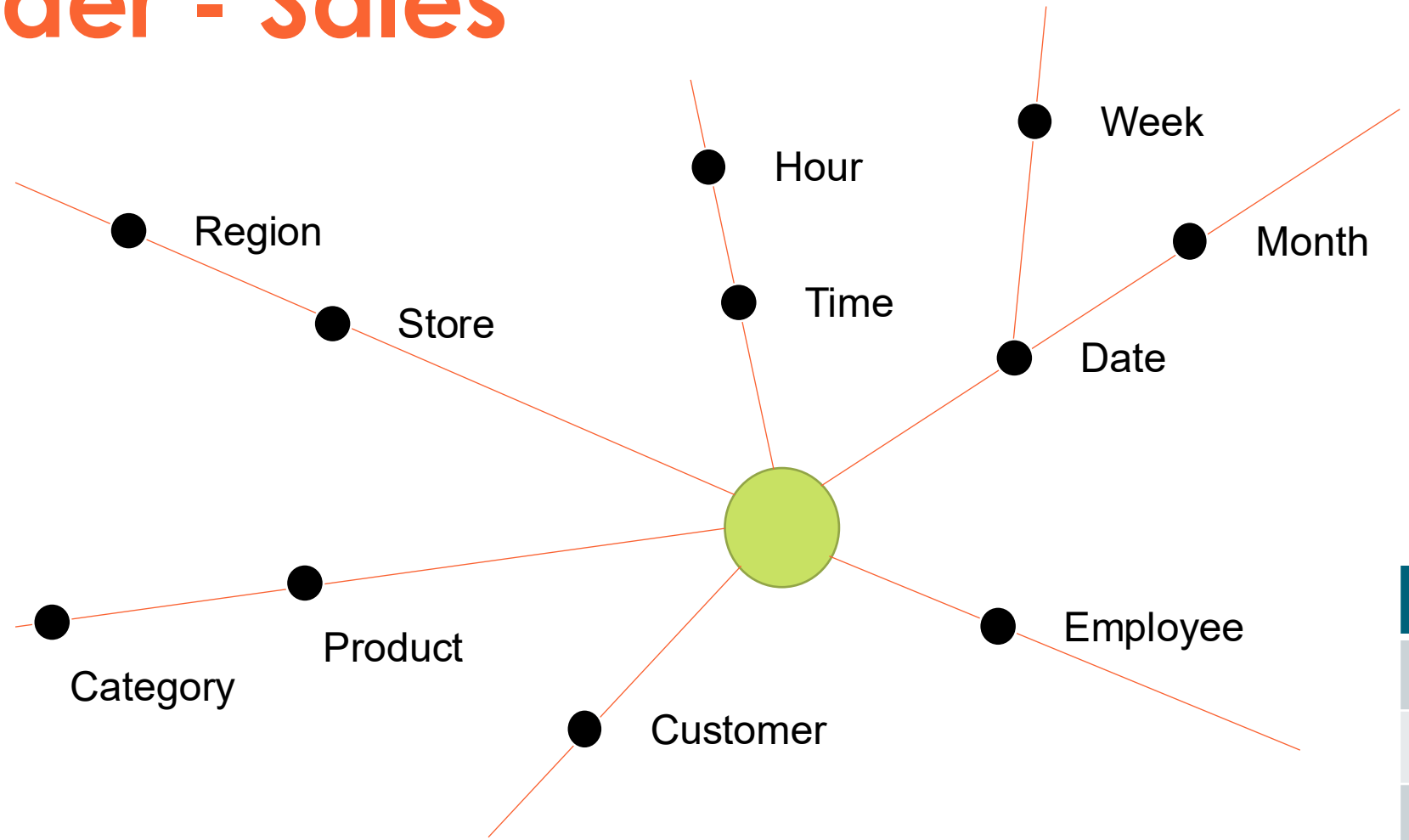

Notice: There no refunds on sale items.
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for exchange.
```

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DIMENSIONAL MODELING

Schema's

Spider - Sales



Facts
Sales Quantity
Sales Amount
Cost Amount
Gross Profit Amount

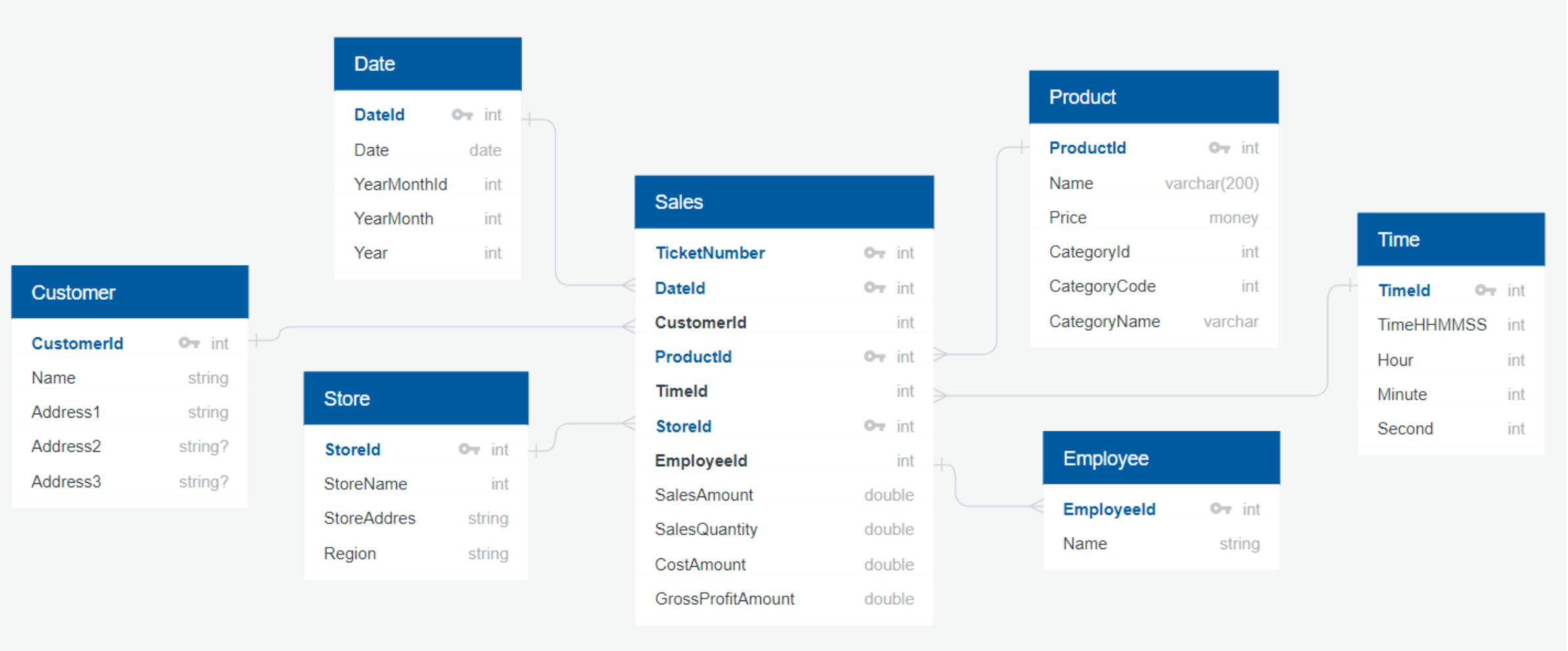
Bus Matrix

		Common / Conformed Dimensions						
		Store	Customer	Product	Employee	Time	Date	Inventory Location
Facts	Sales	X	X	X	X	X	X	

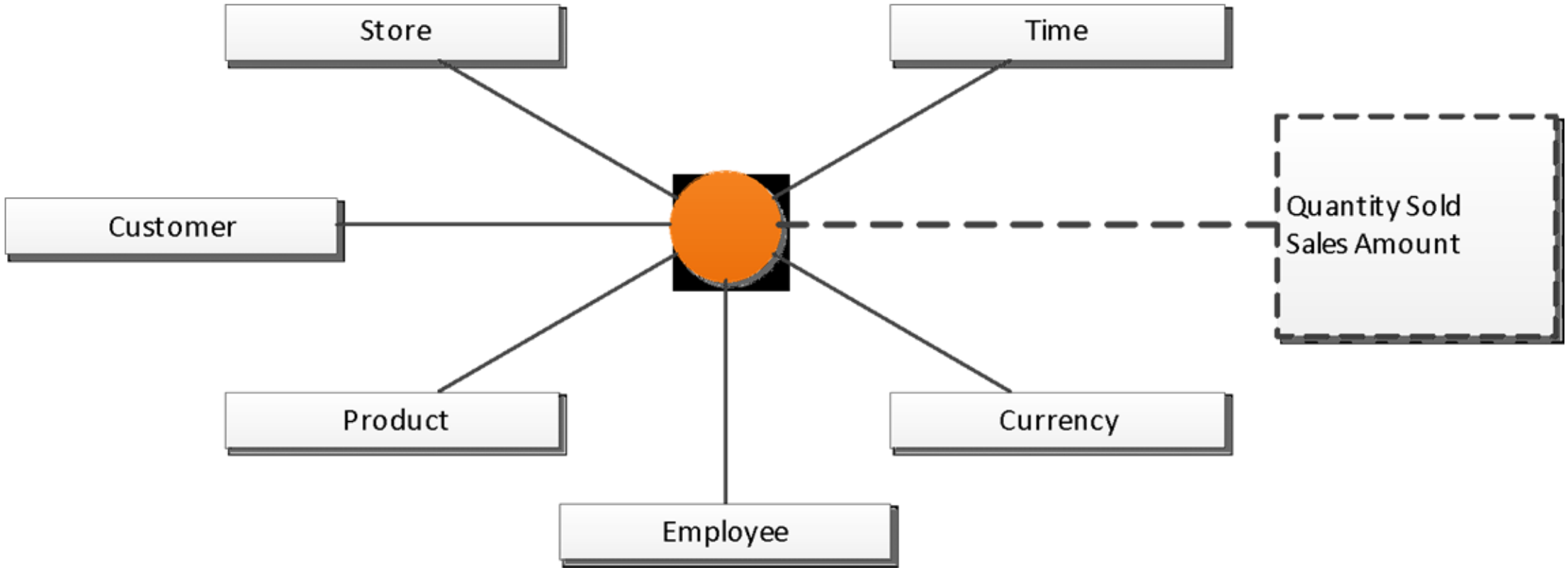
Bus Matrix

Dimensions -> / Subject Areas	Date	Student	Instructor	Employee	Academic Structure	Academic Program	Campus	GL Account	Department	Fund	Project	Course	Award	Degree	Campaign	Proposal	Funding Agency
Student Financials	X	X	X		X	X		X		X		X					
Student Achievement	X	X	X		X	X	X					X	X	X			
Proposals and grants	X			X	X		X		X	X	X					X	X
Admissions	X	X			X		X										
Enrollments	X	X	X		X	X	X					X					
Alumni	X	X													X		
Human Resources	X		X		X				X								
Finance	X				X			X	X								
Course Scheduling	X	X	X		X		X					X					
Recruiting	X	X			X										X		

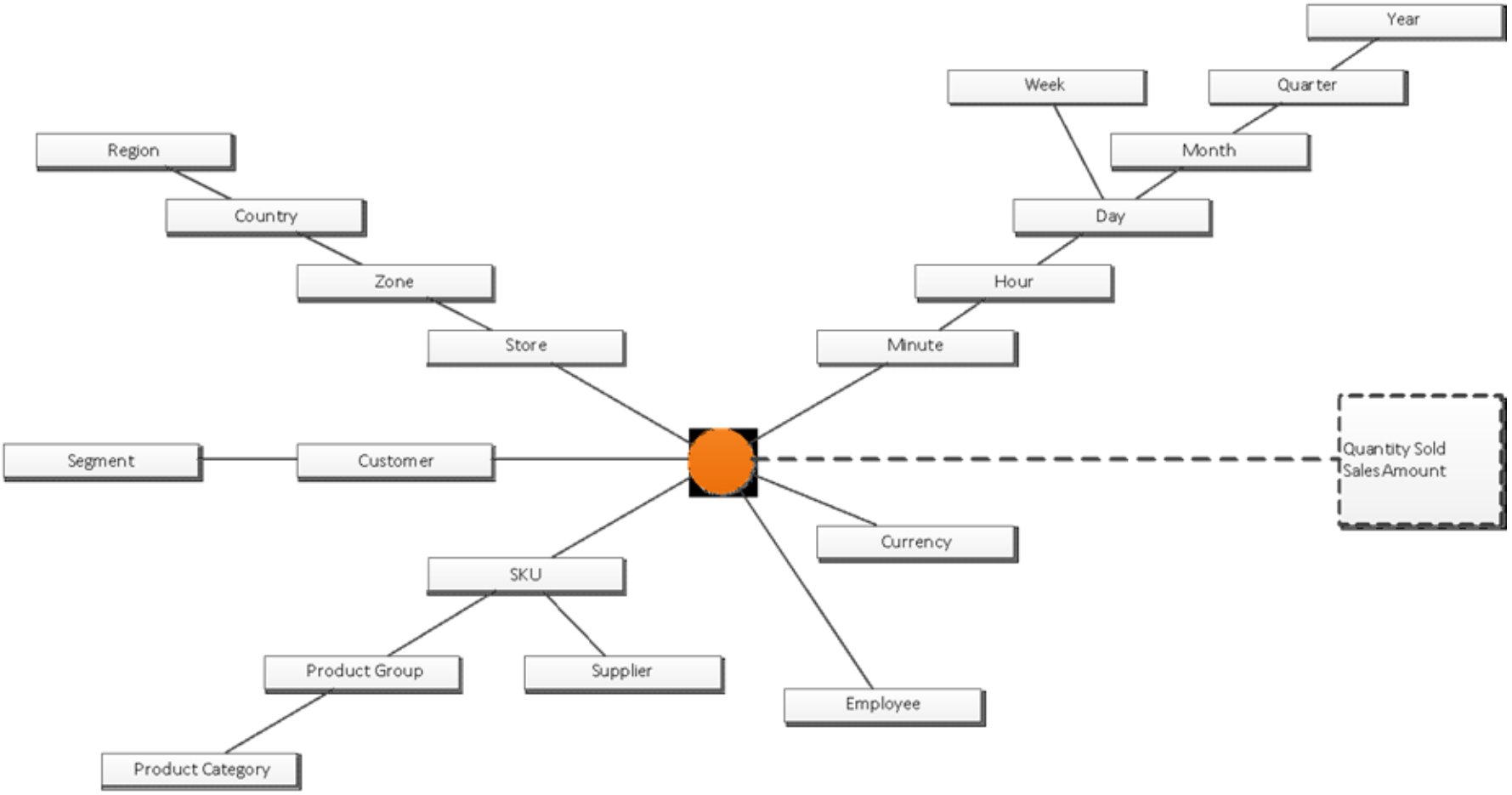
Physical Data Model - Sales



Exercise Business Dimensional Model



Define the hierarchies



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DIMENSIONAL MODELING

Exercise

Case 1: Thomas More airport

At the Thomas More airport, there are a lot of possible indicators that can be measured. In 2024, more than 24 million passengers left via the airport. You can imagine that this generates a big amount of data.

But everything starts with flight data, this is our start.

Can you create a dimensional model that is able to show the base information of a flight.

Case 1: Thomas More airport

Use the 4 steps to define the model
Deliver them all

- Process
- Grain
- Spider
- Bus Matrix
- Physical data model

1. Define the process

1. Define the process

Flightinformation

2. Grain

2. Grain

One flight

3. Identify the dimensions

Carrier
Departure
Arrival
Departure Time/date
Arrival Time/date
Gate
Flightnumber?
Passenger?

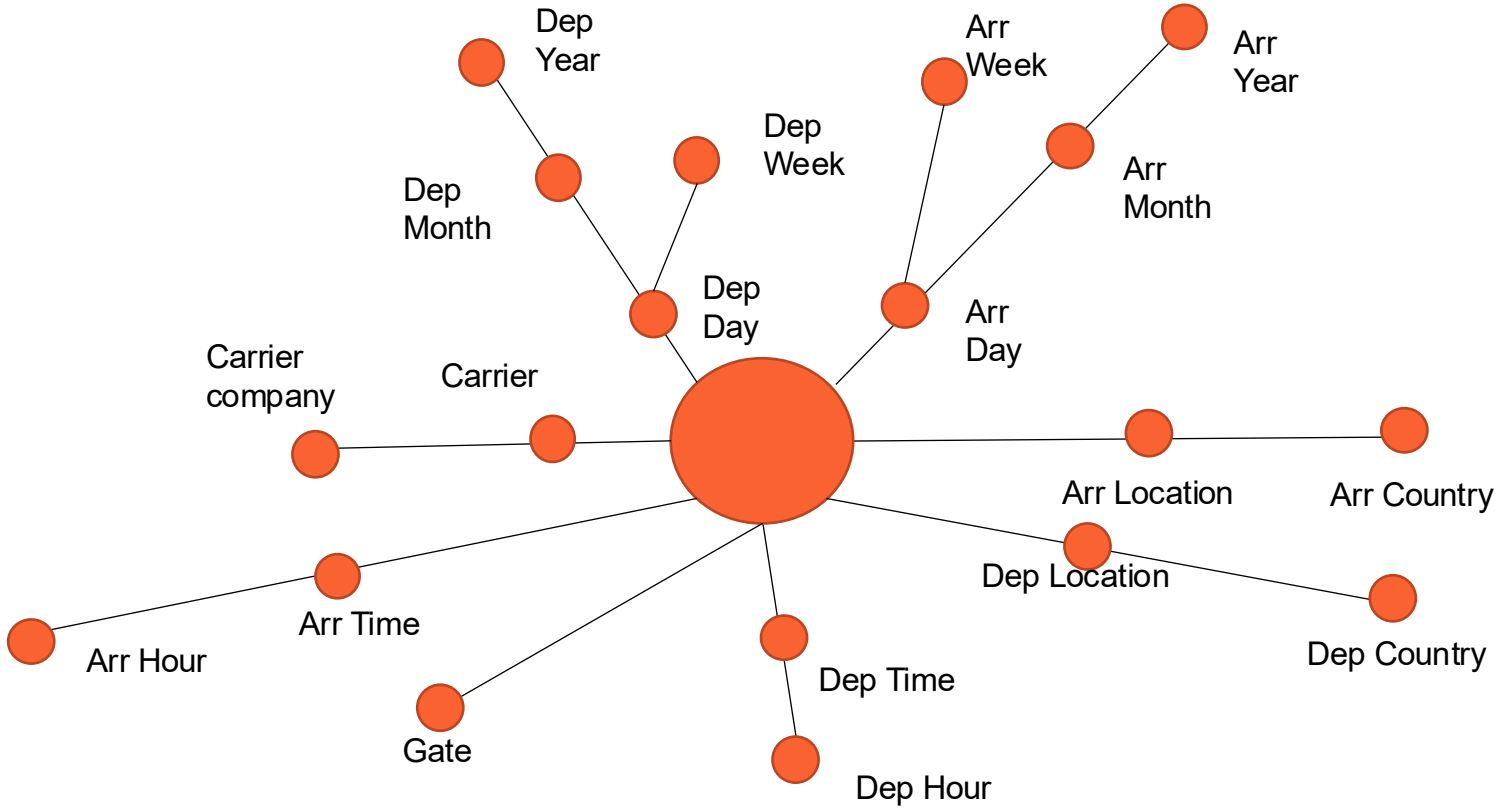
4. Identify the facts

4. Identify the facts

On-time
Amount of flights
Amount of passengers
Difference with schedule (Delaytime)

Spider

Process: Flight



Facts
On-Time
Amount of flights
Amount of pax
Delaytime

Attentionpoints from the flight model

-Level of the fact, always 1-1 with the facttable

-Flights arrive and departs

-2 versions with flag (A/D) or every dimension double

-Flights might have delays

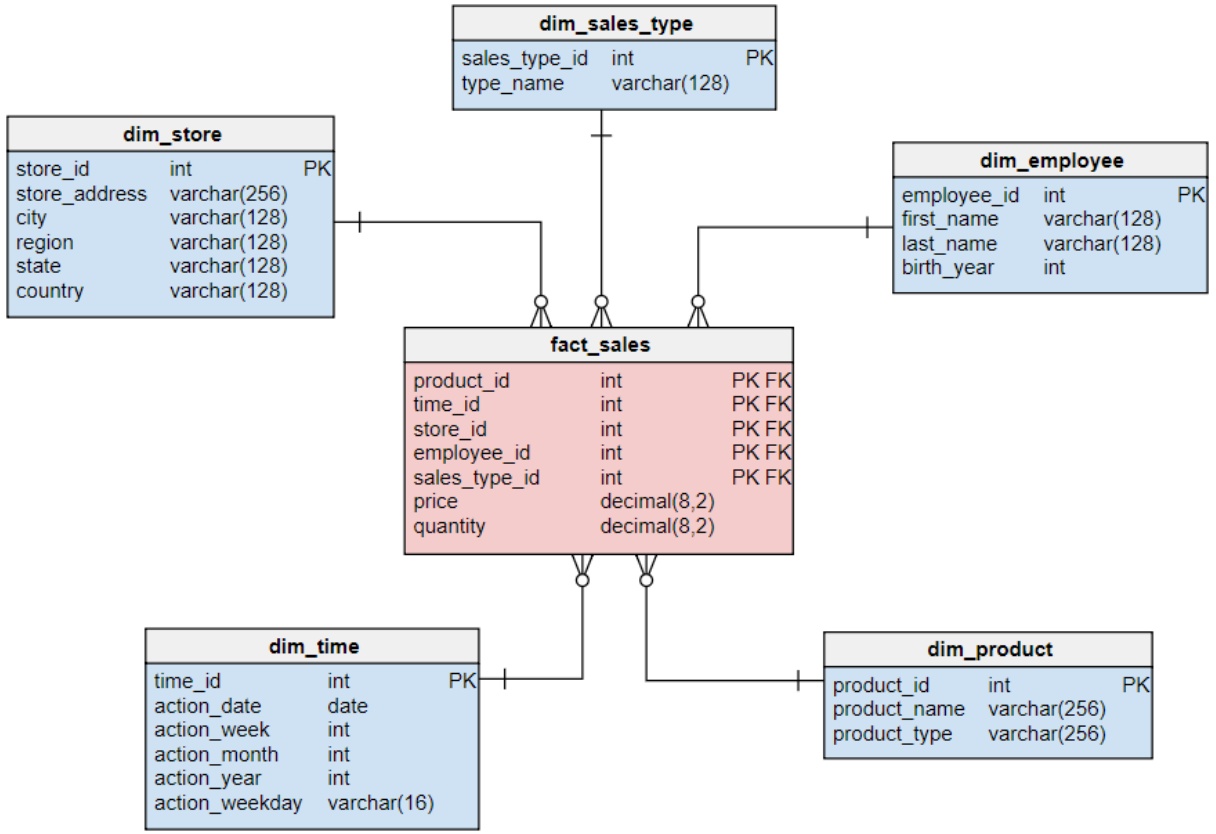
-Spiderschema hierarchy

THOMAS MORE UNIVERSITY OF APPLIED SCIENCES

MULTI-DIMENSIONAL MODELING

Snowflaking

Example starschema



Corresponding SQL Query

```
SELECT P.Brand, S.Country, SUM (F.Units_Sold)
 FROM Fact_Sales F
 INNER JOIN Dim_Date D ON F.Date_Id = D.Id
 INNER JOIN Dim_Store S ON F.Store_Id = S.Id
 INNER JOIN Dim_Product P ON F.Product_Id = P.Id
WHERE D.Year = 2015
AND P.Product_Category = 'tv'
GROUP BY P.Brand, S.Country
```

Dimensional datamodel: Snowflake

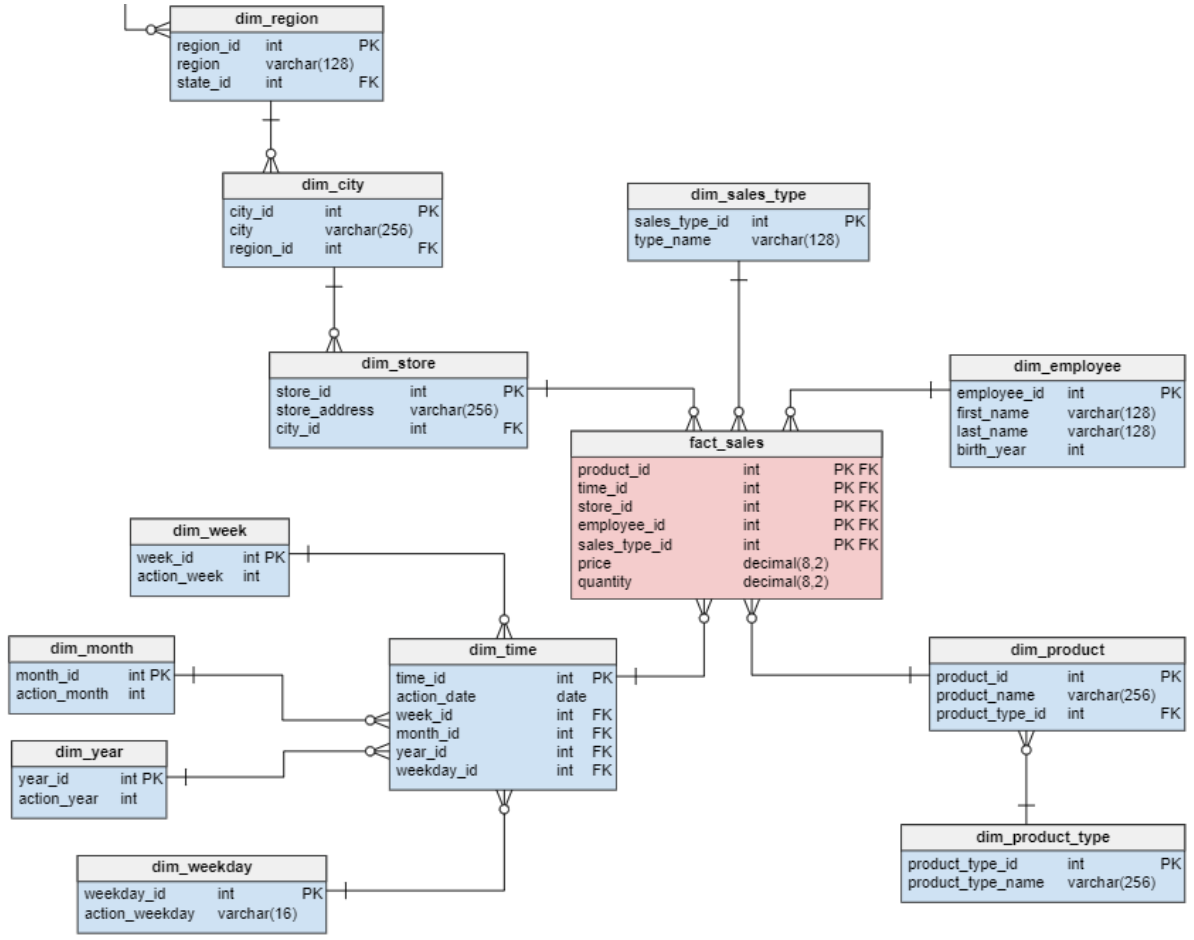
A Snowflake schema is a starschema where 1 or many dimensiontables are normalized

Less redundant information

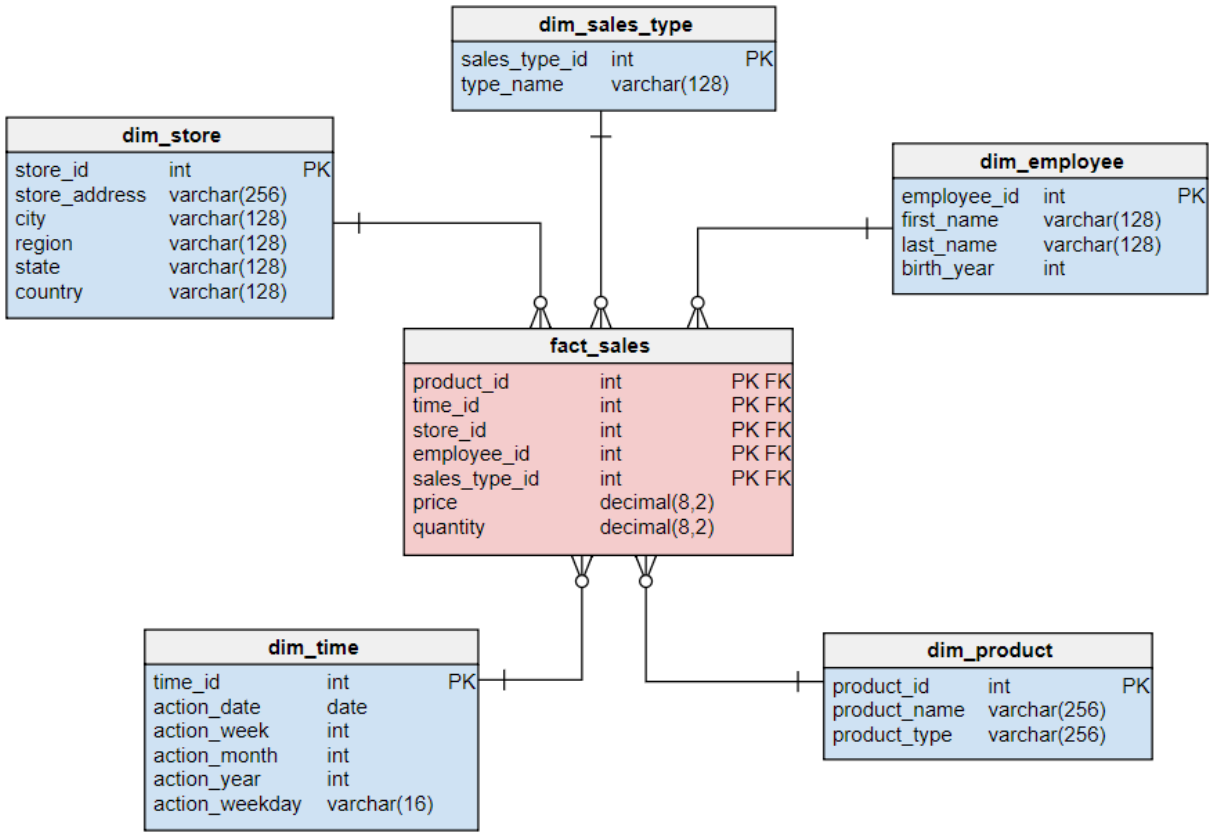
More relations between the entities, this causes many JOINS with queries. This impacts the performance in a negative way

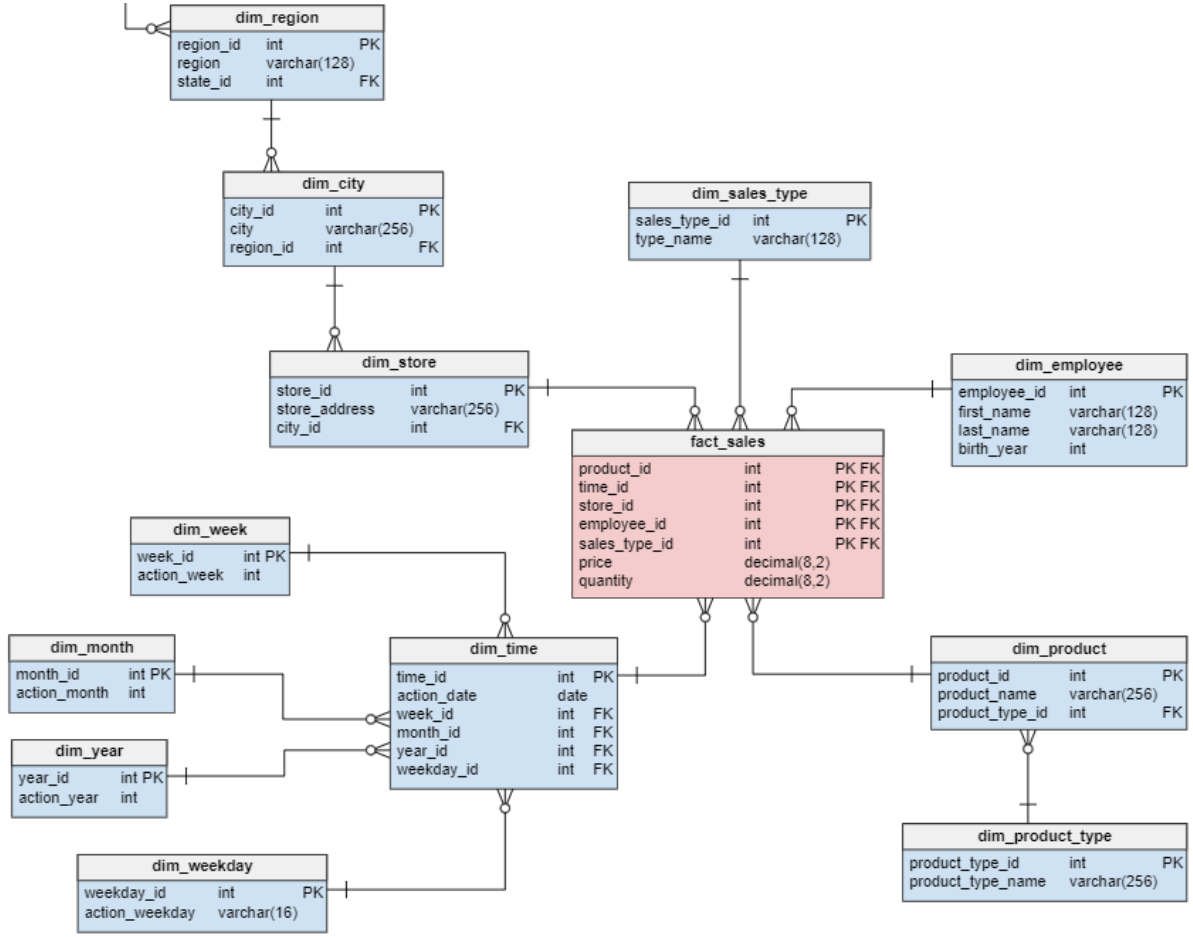
Snowflake

Every hierarchy in a dimensiontable wil be declared in a sepearte dimensiontable



Star vs Snowflake schema



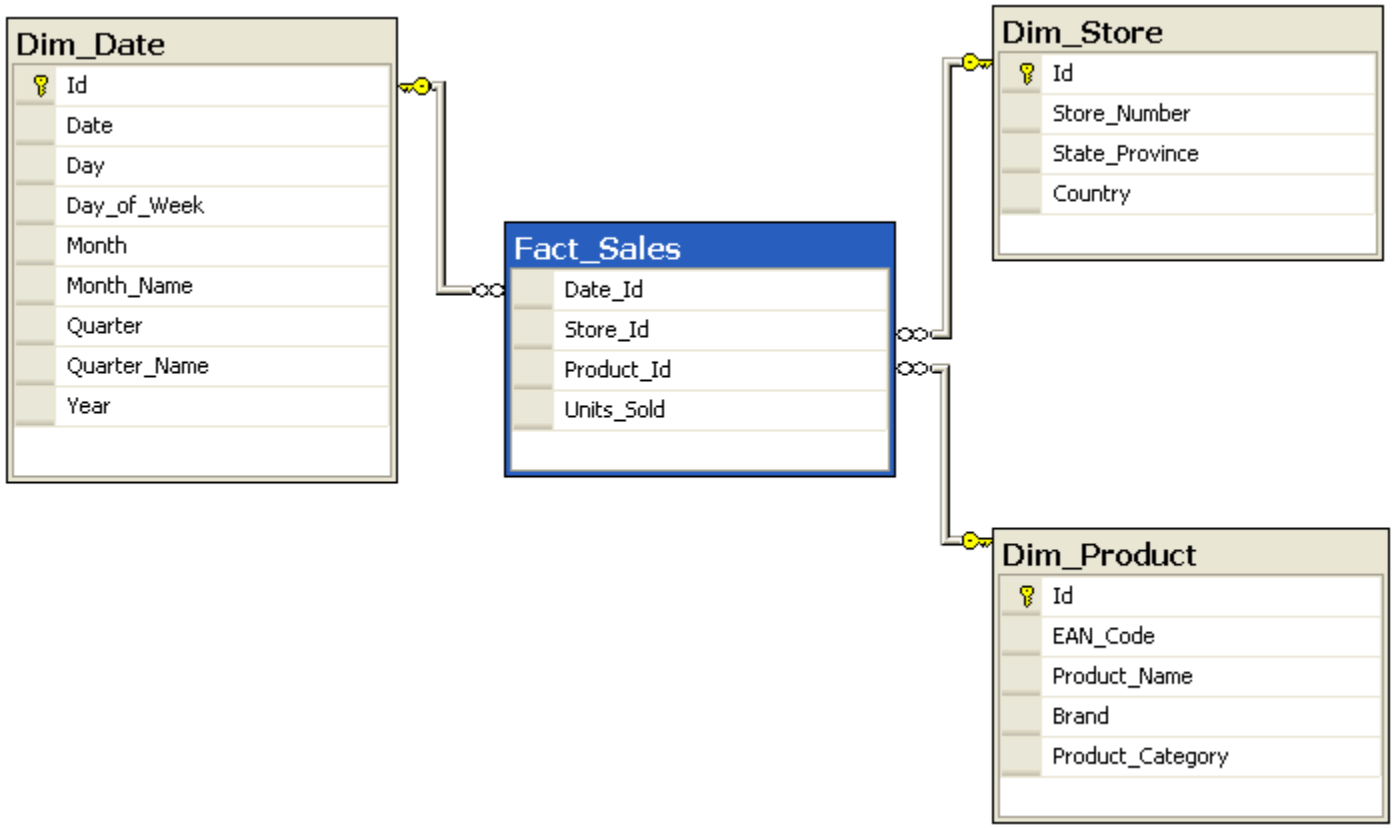


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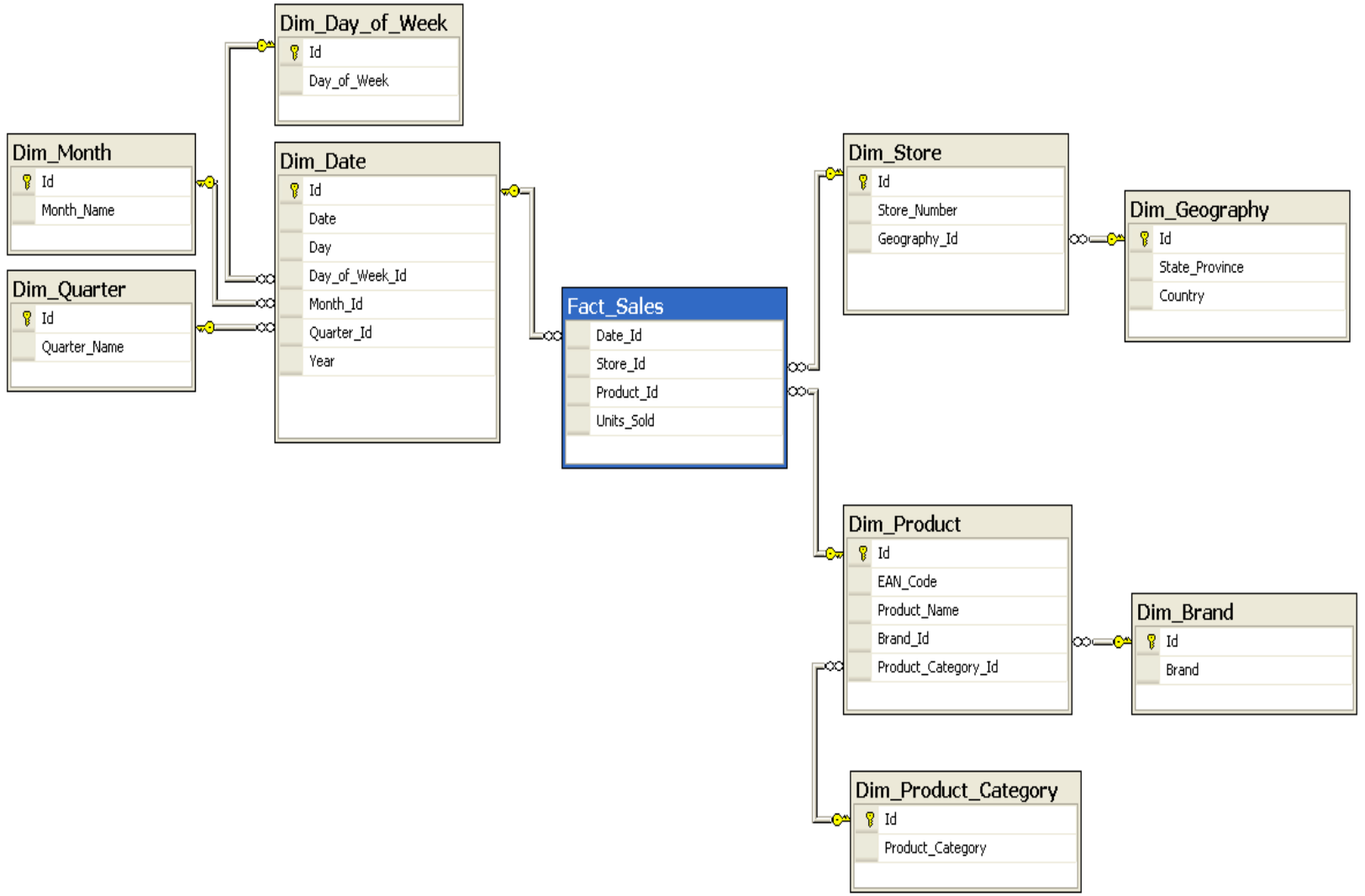
MULTI-DIMENSIONAL MODELING

Snowflaking exercise

Example starschema



Example snowflake



Corresponding SQL query

```
SELECT B.Brand, G.Country, SUM (F.Units_Sold)
FROM Fact_Sales F
INNER JOIN Dim_Date D ON F.Date_Id = D.Id
INNER JOIN Dim_Store S ON F.Store_Id = S.Id
INNER JOIN Dim_Geography G ON S.Geography_Id = G.Id
INNER JOIN Dim_Product P ON F.Product_Id = P.Id
INNER JOIN Dim_Product_Category C ON P.Product_Category_Id = C.Id
INNER JOIN Dim_Brand B ON P.Brand_Id = B.Id

WHERE D.Year = 2013 AND C.Product_Category = 'tv'
GROUP BY B.Brand, G.Country
```

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MULTI-DIMENSIONAL MODELING

Multiple facts

Ticketline



```
      HOTEL UNIVERSAL RESORT
       RJ SHIPS AND GIFT STORE


11/11/15 10:21am     Ticket No: 110595
----------------------------------------
T:01 Shft:1 Cshr:829         Guests:
Salesprsn:SMITH
----------------------------------------
ICE                                 3.00
COOLER 4 GAL                        8.95
ARIZONA TEA       2.00 @  2.75      5.50
NECKACE SILVER/ BLUE               25.00
            10% DISCOUNT           (2.50)
PURSES (18.00)                     18.00
                                 ----------
                Sub-total...:      60.45
                Discount            2.50-
                TAX                 4.35
                                 ----------

                      Total:       62.30

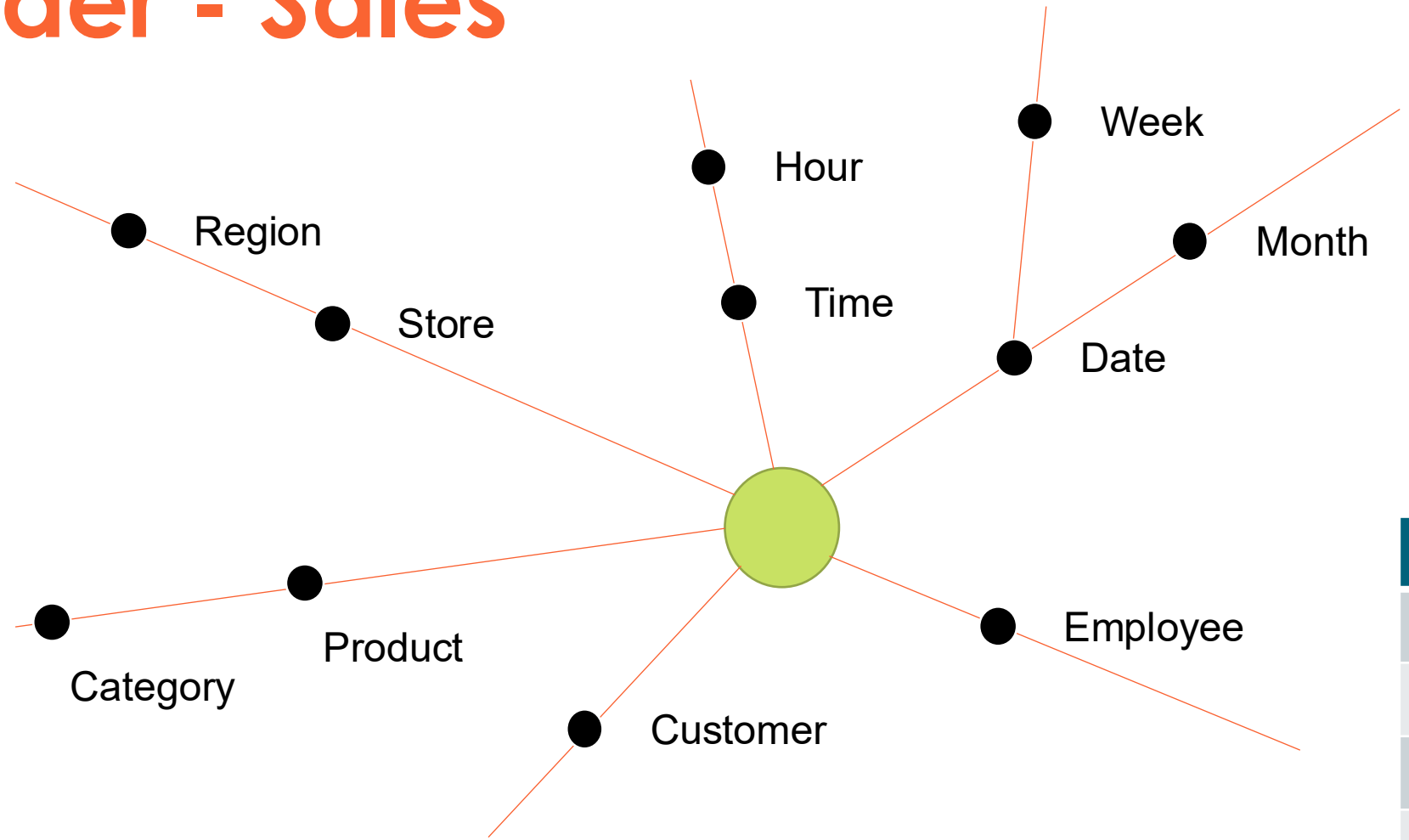

PAYMENT- CASH


THANK YOU



Notice: There no refunds on sale items.
Receipt/proof of purchase is required
for exchange.
```

Spider - Sales

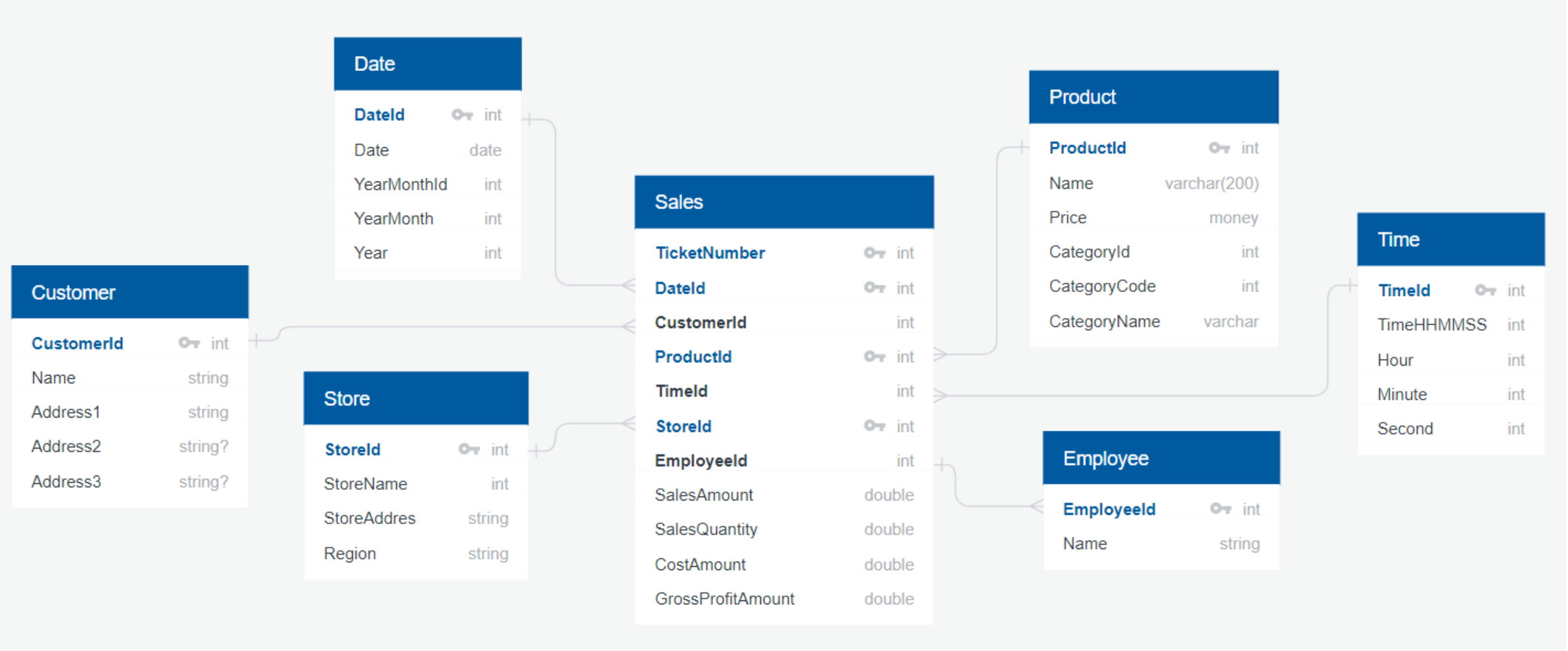


Facts
Sales Quantity
Sales Amount
Cost Amount
Gross Profit Amount

Bus Matrix

		Common / Conformed Dimensions					
		Store	Customer	Product	Employee	Time	Date
Facts	Sales	X	X	X	X	X	X

Physical Data Model - Sales



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DIMENSIONAL MODELING

Exercise budget

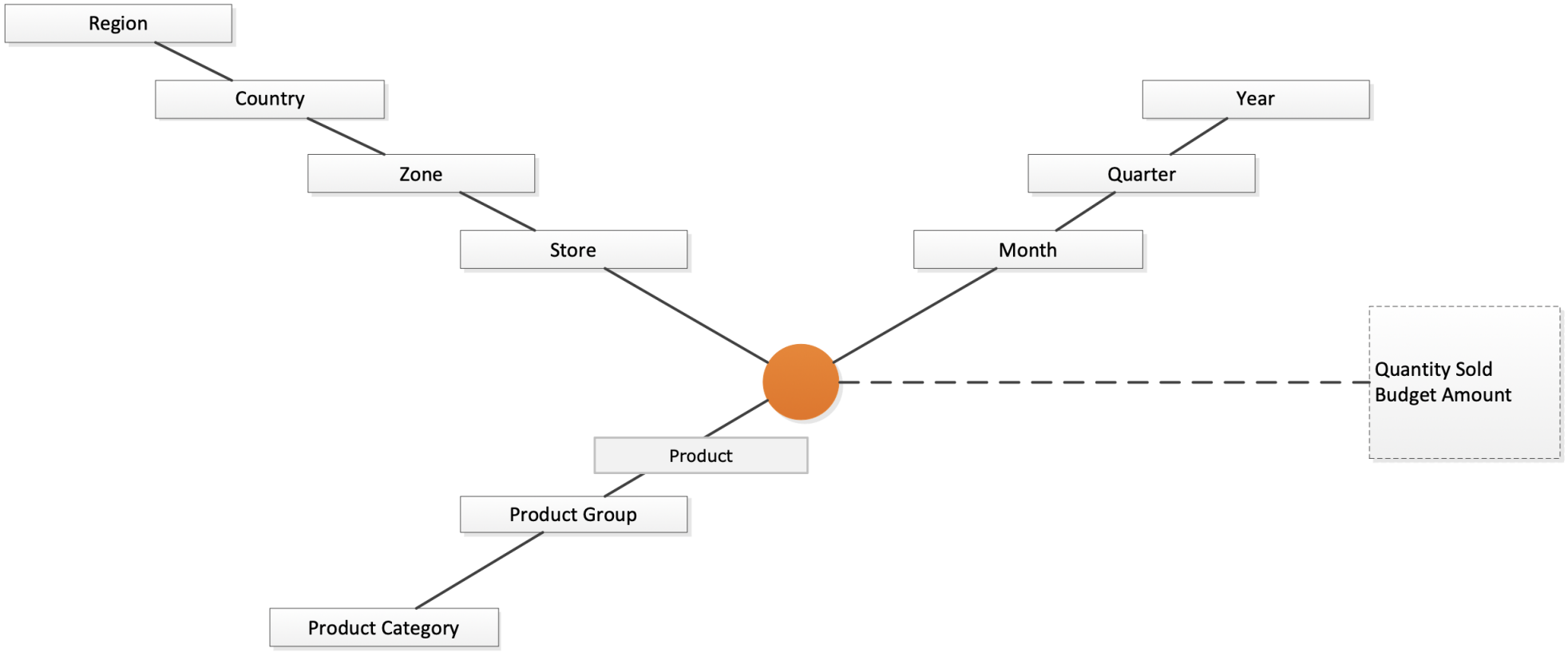
Add Budget

Create a budgetmodel to

- Compare budget vs sales

Spider + bus matrix + physical model

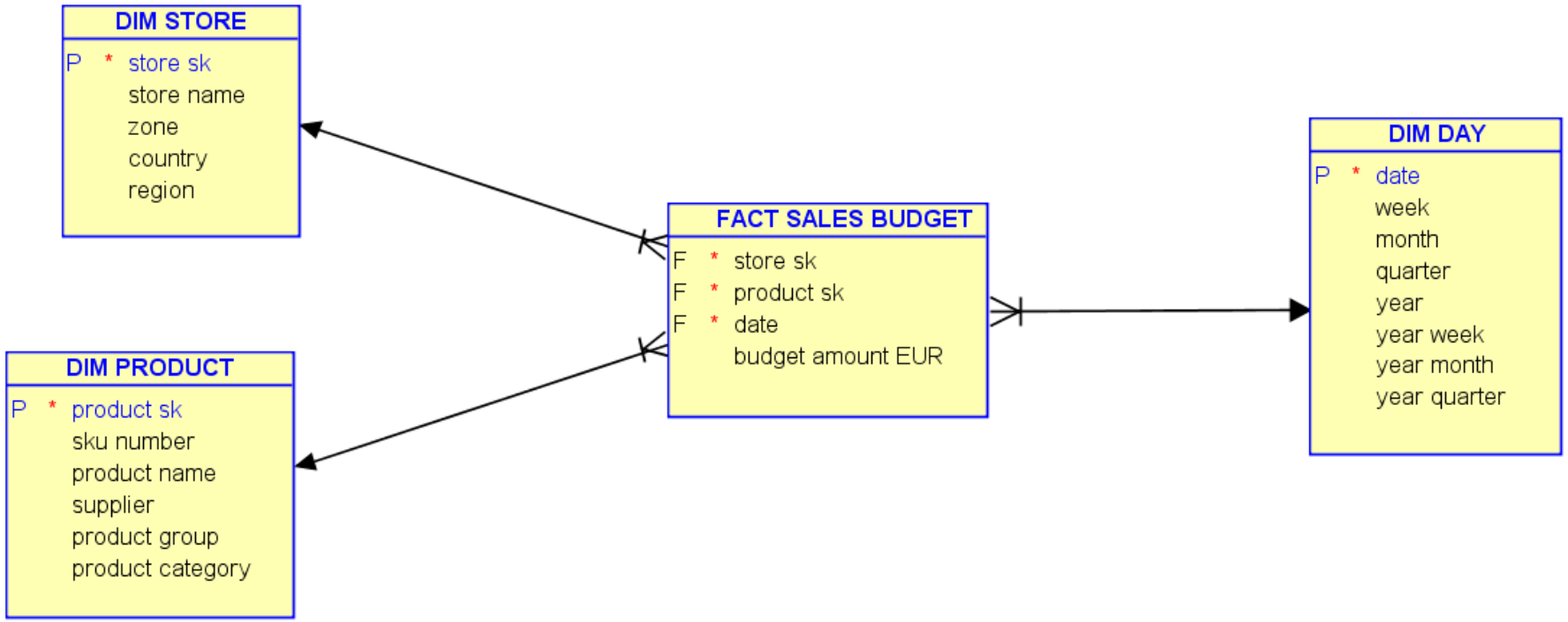
Business Dimensional Model



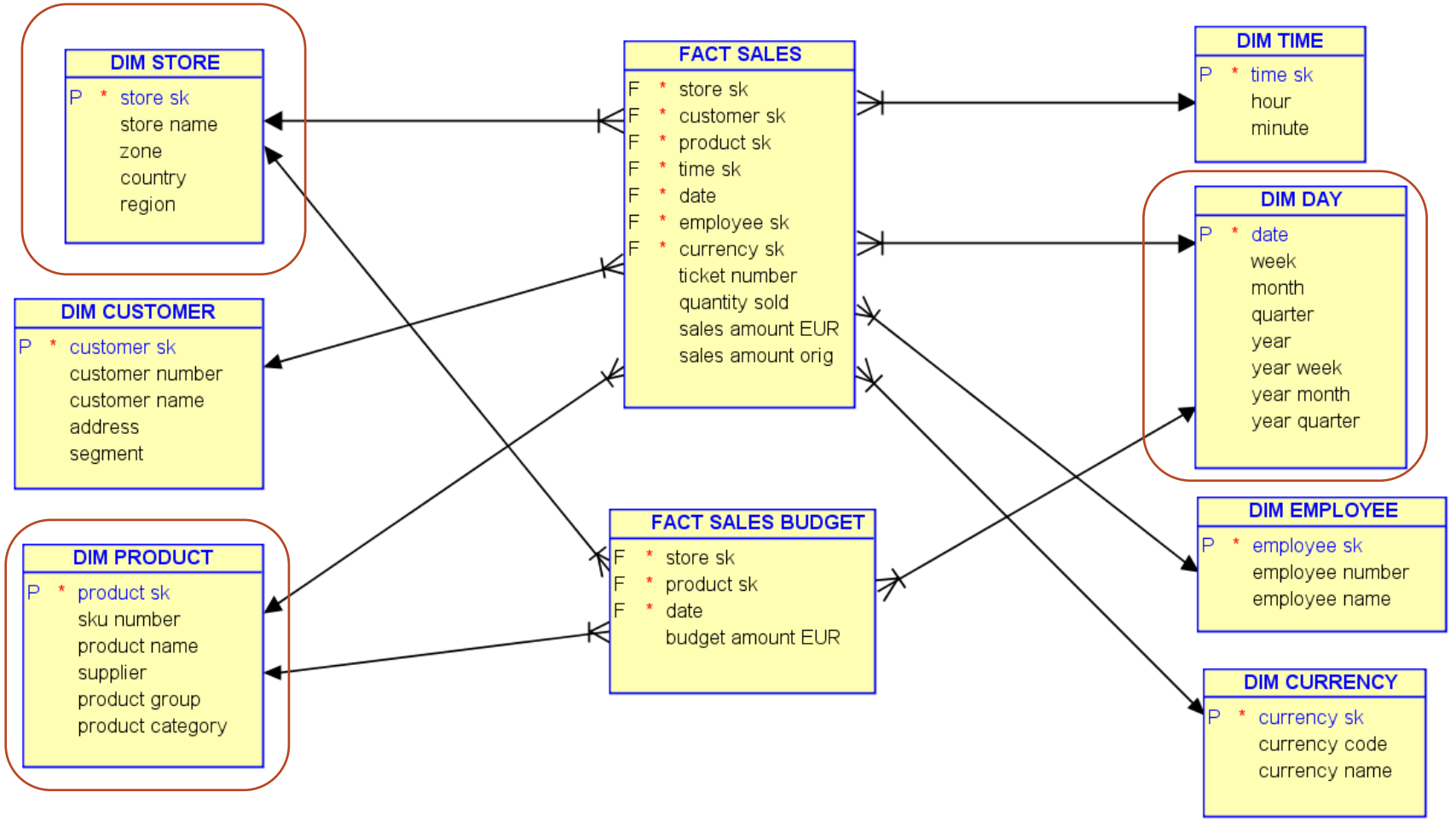
Enterprise Bus Matrix

		Common / Conformed Dimensions							
		Store	Customer	Product	Promotion	Employee	Currency	Time	Date
Facts	Sales	X	X	X	X	X	X	X	X
	Sales Budget	X		X					X

Sales Budget model



Integrated model



Conformed Dimensions

Shared by multiple facts
Consistency
Maintained in 1 place which causes less development
Every model uses the same structure
Combine multiple facts in 1 query

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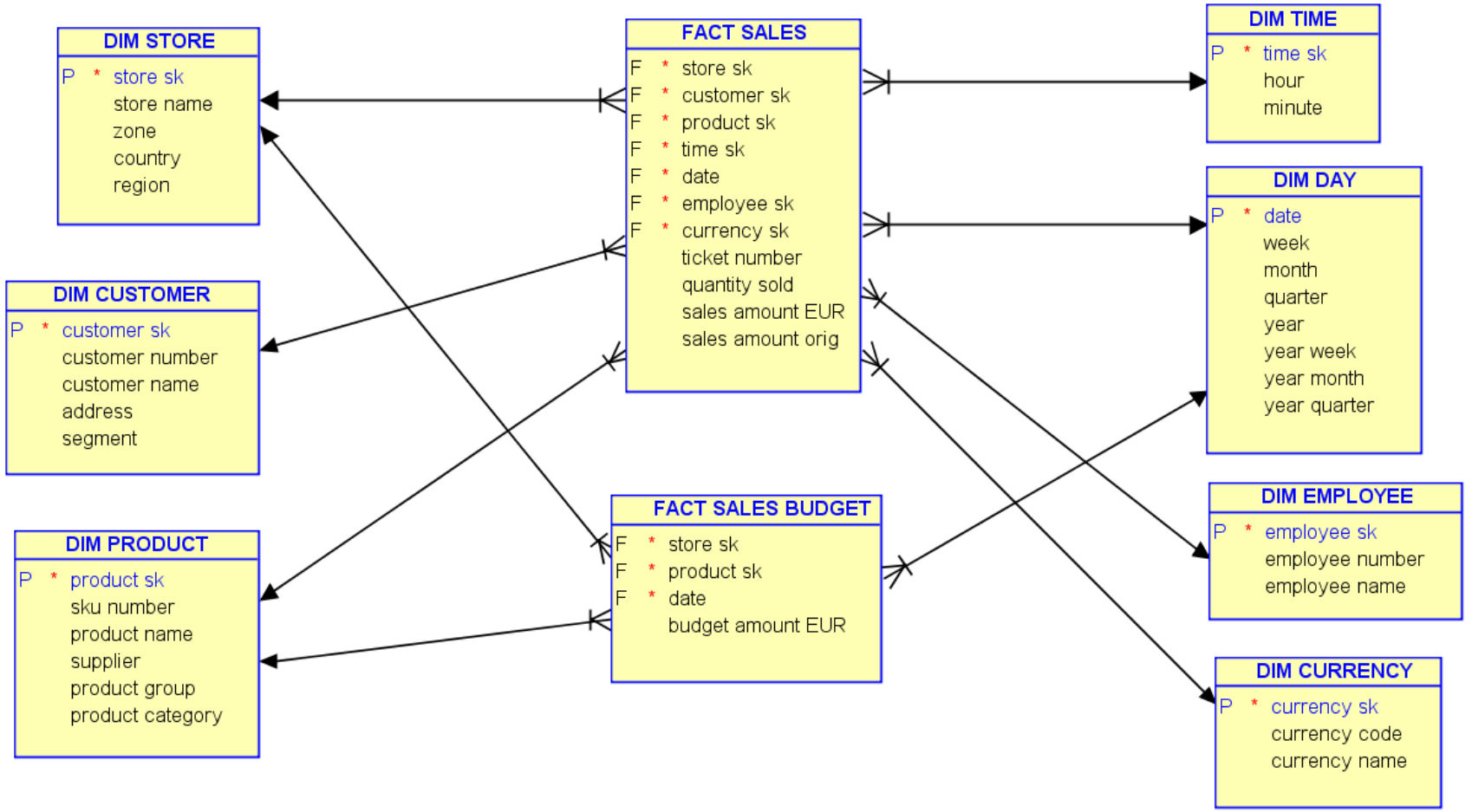
DIMENSIONAL MODELING

Exercise promotions

Promotion analysis

Change the current solution to create a model that also allows promotion analysis

Previous solution



Promotions

```
      HOTEL UNIVERSAL RESORT
       RJ SHIPS AND GIFT STORE


11/11/15 10:21am     Ticket No: 110595
----------------------------------------
T:01 Shft:1 Cshr:829         Guests:
Salesprsn:SMITH
----------------------------------------
ICE                                 3.00
COOLER 4 GAL                        8.95
ARIZONA TEA       2.00 @  2.75      5.50
NECKACE SILVER/ BLUE               25.00
            10% DISCOUNT           (2.50)
PURSES (18.00)                     18.00
                                 ----------
                Sub-total...:      60.45
                Discount            2.50-
                TAX                 4.35
                                 ----------

                      Total:       62.30

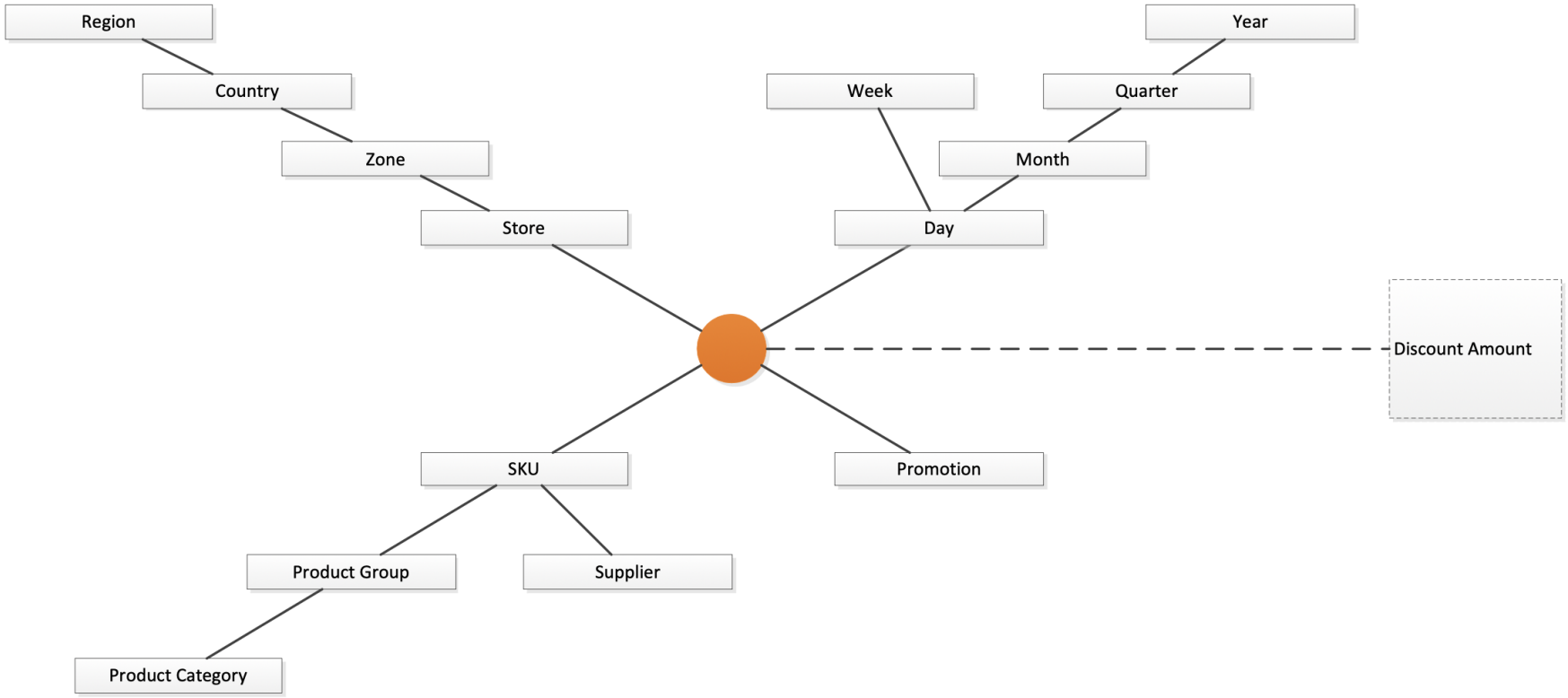

PAYMENT- CASH


THANK YOU

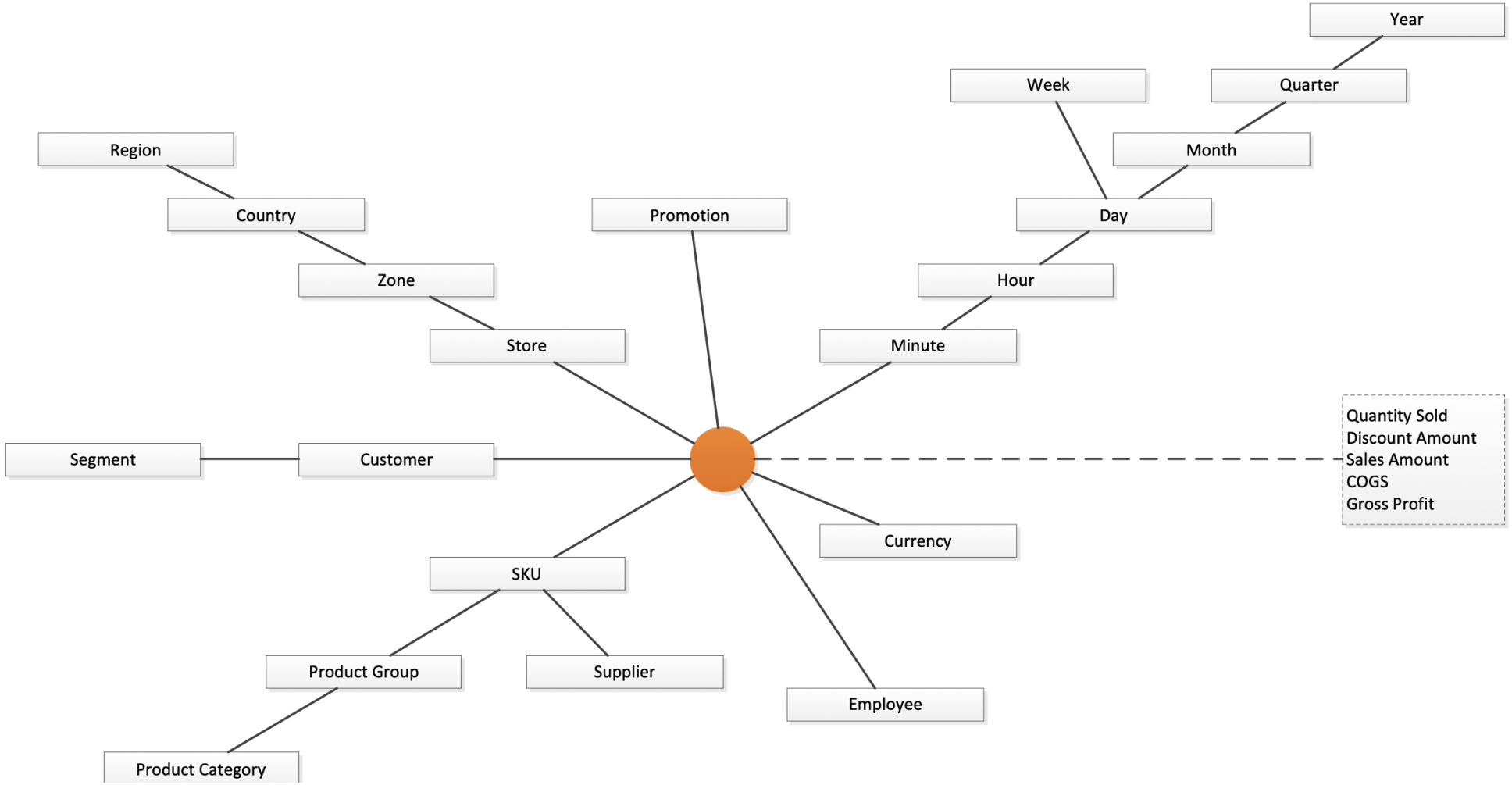


Notice: There no refunds on sale items.
Receipt/proof of purchase is required
for exchange.
```

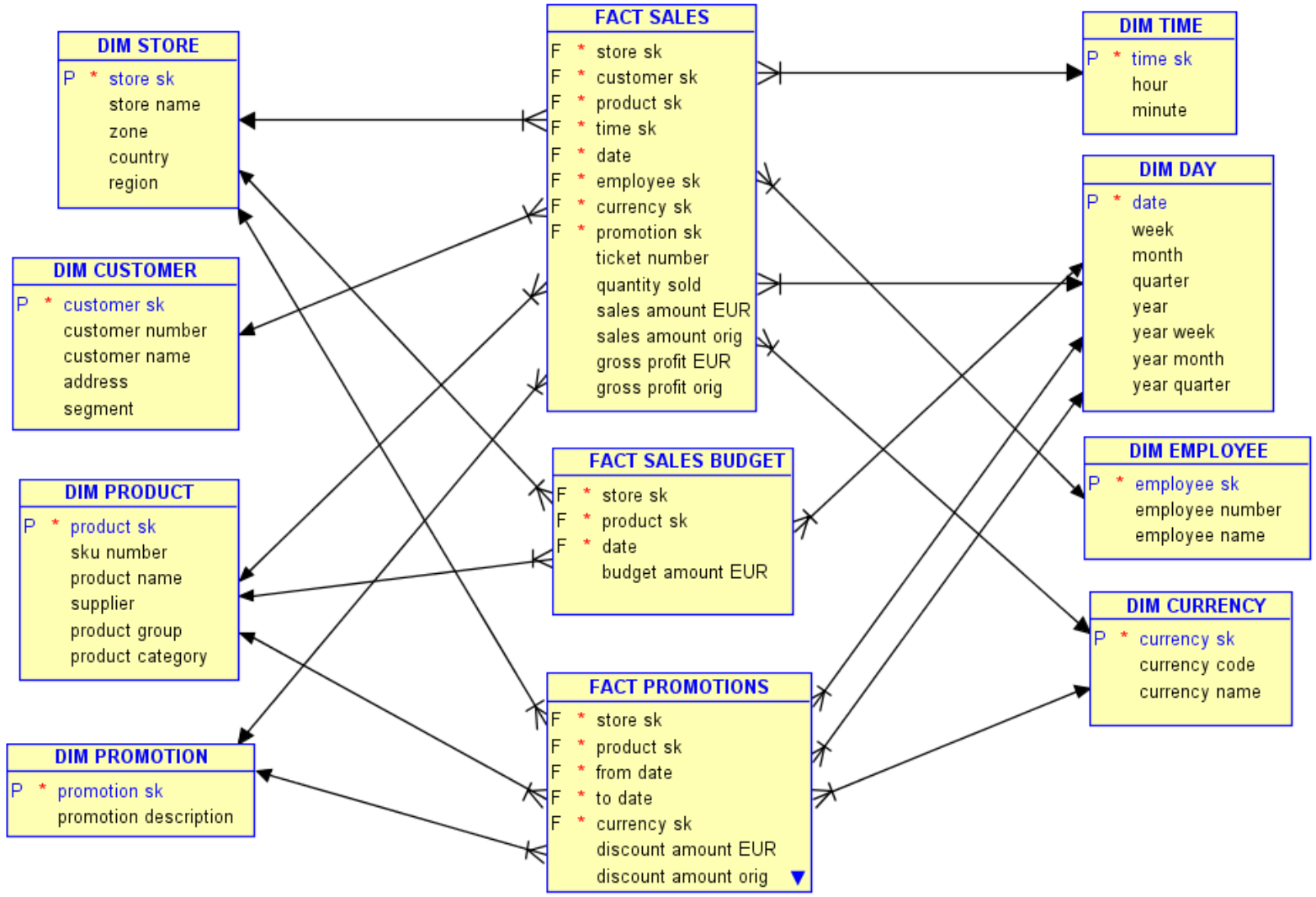
Only promotions



Including promotions



Datamodel


DIM STORE
P * store sk
store name
zone
country
region
DIM CUSTOMER
P * customer sk
customer number
customer name
address
segment
DIM PRODUCT
P * product sk
sku number
product name
supplier
product group
product category
DIM PROMOTION
P * promotion sk
promotion description
FACT SALES
F * store sk
F * customer sk
F * product sk
F * time sk
F * date
F * employee sk
F * currency sk
F * promotion sk
ticket number
quantity sold
sales amount EUR
sales amount orig
gross profit EUR
gross profit orig
FACT SALES BUDGET
F * store sk
F * product sk
F * date
budget amount EUR
FACT PROMOTIONS
F * store sk
F * product sk
F * from date
F * to date
F * currency sk
discount amount EUR
discount amount orig
DIM TIME
P * time sk
hour
minute
DIM DAY
P * date
week
month
quarter
year
year week
year month
year quarter
DIM EMPLOYEE
P * employee sk
employee number
employee name
DIM CURRENCY
P * currency sk
currency code
currency name

Enterprise Bus Matrix

		Common / Conformed Dimensions							
		Store	Customer	Product	Promotion	Employee	Currency	Time	Date
Facts	Sales	X	X	X	X	X	X	X	X
	Sales Budget	X		X					X
	Promotions	X		X	X		X		X

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DIMENSIONAL MODELING

Extra exercise lesson 2

Case 2: Thomas More airport

Now that all the flights of the Thomas More Airport are embedded in our system, it's time to expand our analysises. An airport isn't the same as a carrier and generates the most of their profit out of real-estate and commercial activities. With real-estate we mean renting of buildings to carries for their offices, cargo, hangars to park plains,..
But the most profit they generate our of shops.

Create a model that allows analysis on the sales in the shops at the airport

Case 2: Sales within the Thomas More airport

Use the 4 steps to define the model
Deliver:

- Grain
- Spider
- Bus Matrix
- physical data model

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DIMENSIONAL MODELING

Slowly changing dimensions

SCD

Slowly Changing Dimensions

- Attributes changes over time
  - New productname, address, …

- What do you need to do with the data when someone moves from Belgium to The Netherlands, where do you store the results in Belgium or The Netherlands?

SCD

Customer_ID	FirstName	LastName	Customer_type
100	Steve	Hillen	Retail
101	Jef	De Wit	Corporate

Customer_ID	FirstName	LastName	Customer_type
100	Steve	Hillen	Corporate
101	Jef	De Wit	Corporate

SCD Type 1

3 ways to handle Slowly Changing Dimensions

Type 1SCD: Overwrite old data with new

- Historical data is wrong

Customer_ID	FirstName	LastName	Customer_type
100	Steve	Hillen	Retail
101	Jef	De Wit	Corporate

Customer_ID	FirstName	LastName	Customer_type
100	Steve	Hillen	Corporate
101	Jef	De Wit	Corporate

SCD Type 1

Used for

- Data Corrections (birthdates etc)
- Overwrite existing values
- No Historical Values

SCD Type 2

3 ways to handle Slowly Changing Dimensions

Type 2 SCD: add new record in dimensiontable

- Historical data remains correct in the database
- You receive up to data records and the closed records are closed by adding extra attributes like DateStart, DateStop
- Foresee a way to count records in the past and now (old and new records).

SCD Type 2

Example Type 2 SCD:
Add new records into the dimension

Customer_ID	FirstName	LastName	Customer_type	Start_date	End_date	Current_flag
100	Steve	Hillen	Retail	01/01/2019	31/08/2021	N
101	Jef	De Wit	Corporate	01/05/2019	31/12/9999	Y
100	Steve	Hillen	Corporate	01/09/2021	31/12/9999	Y

→ Link between 2 records is lost!

→ Problem to count records!

how do you count the rows for Steve?

SCD Type 2

Example Type 2 SCD:
add new record to the dimension with a combined PK key

Customer_ID	FirstName	LastName	Customer_type	Start_date	End_date	Current_flag
100	Steve	Hillen	Retail	01/01/2019	31/08/2021	N
101	Jef	De Wit	Corporate	01/05/2019	31/12/9999	Y
100	Steve	Hillen	Corporate	01/09/2021	31/12/9999	Y

SCD Type 2

Example Type 2 SCD:
add new record in the dimension
add a new ID

ID	Customer_ID	FirstName	LastName	Customer_type	Start_date	End_date	Current_flag
1	100	Steve	Hillen	Retail	01/01/2019	31/08/2021	N
2	101	Jef	De Wit	Corporate	01/05/2019	31/12/9999	Y
3	100	Steve	Hillen	Corporate	01/09/2021	31/12/9999	Y

SCD Type 2

Track all history changes
Create a new record with date_start and date_stop
More rows

3. Define Dimensions

- Type 3 SCD: add attribute with new value
  - Store a attribute in 2 columns, 1 with the old value and 1 with the new value.
  - A last change date will be added

Customer_ID	FirstName	LastName	Customer_type_new	Customer_type_old	Start_date
100	Steve	Hillen	Corporate	Retail	01/01/2019
101	Jef	De Wit	Corporate	Corporate	01/05/2019

SCD type 0 to 7

SCD Type	Dimension Table Action	Impact on Fact Analysis
Type 0	No change to attribute value	Facts associated with attribute's original value
Type 1	Overwrite attribute value	Facts associated with attribute's current value
Type 2	Add new dimension row for profile with new attribute value	Facts associated with attribute value in effect when fact occurred
Type 3	Add new column to preserve attribute's current and prior values	Facts associated with both current and prior attribute alternative values
Type 4	Add mini-dimension table containing rapidly changing attributes	Facts associated with rapidly changing attributes in effect when fact occurred
Type 5	Add type 4 mini-dimension, along with overwritten type 1 mini-dimension key in base dimension	Facts associated with rapidly changing attributes in effect when fact occurred, plus current rapidly changing attribute values
Type 6	Add type 1 overwritten attributes to type 2 dimension row, and overwrite all prior dimension rows	Facts associated with attribute value in effect when fact occurred, plus current values
Type 7	Add type 2 dimension row with new attribute value, plus view limited to current rows and/or attribute values	Facts associated with attribute value in effect when fact occurred, plus current values

SCD

According to Kimball, there are 8 ways to handle SCD, from type 0 until 7
More information:
http://www.kimballgroup.com/data-warehouse-business-intelligence-resources/kimball-techniques/dimensional-modeling-techniques/

Surrogate Key

A sequentially generated unique number
There's no source dependency
Increase performance in linking on ID's compared with text

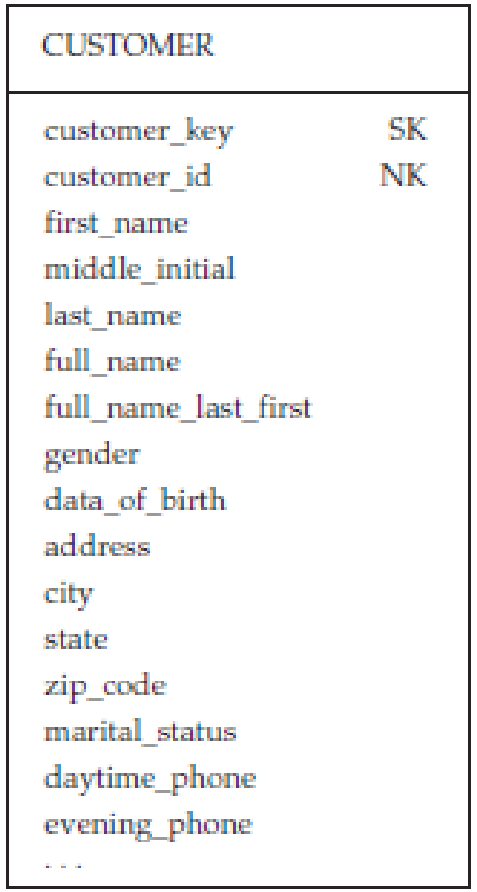
ID	Customer_ID	FirstName	LastName	Customer_type	Start_date	End_date	Current_flag
1	100	Steve	Hillen	Retail	01/01/2019	31/08/2021	N
2	101	Jef	De Wit	Corporate	01/05/2019	31/12/9999	Y
3	100	Steve	Hillen	Corporate	01/09/2021	31/12/9999	Y

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DIMENSIONAL MODELING

Exercise Lesson 3: choose SCD model

exercise


CUSTOMER
customer_key SK
customer_id NK
first_name
middle_initial
last_name
full_name
full_name_last_first
gender
data_of_birth
address
city
state
zip_code
marital_status
daytime_phone
evening_phone
. . .

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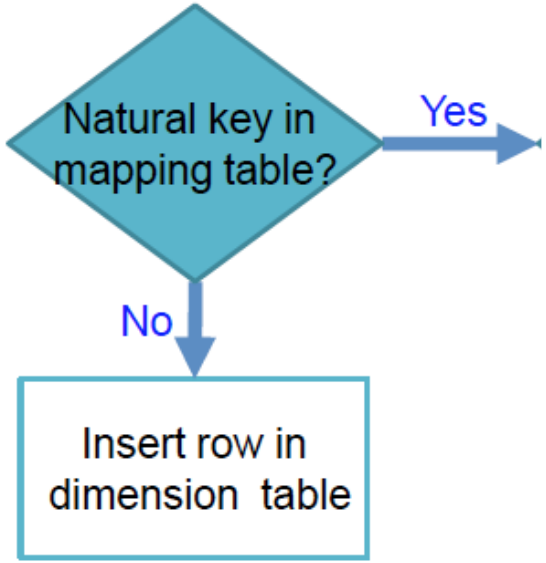
DIMENSIONAL MODELING

Loading dimensions

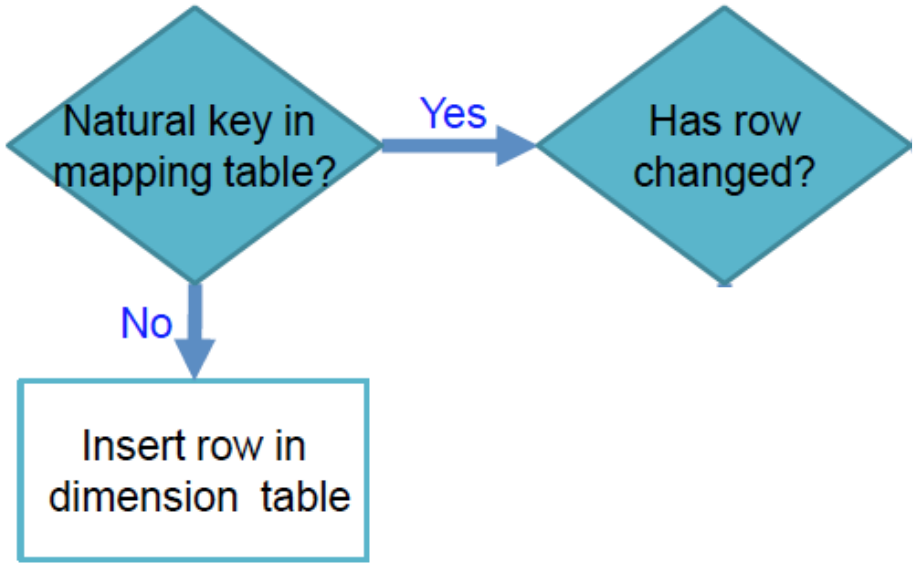
Loading of dimensions



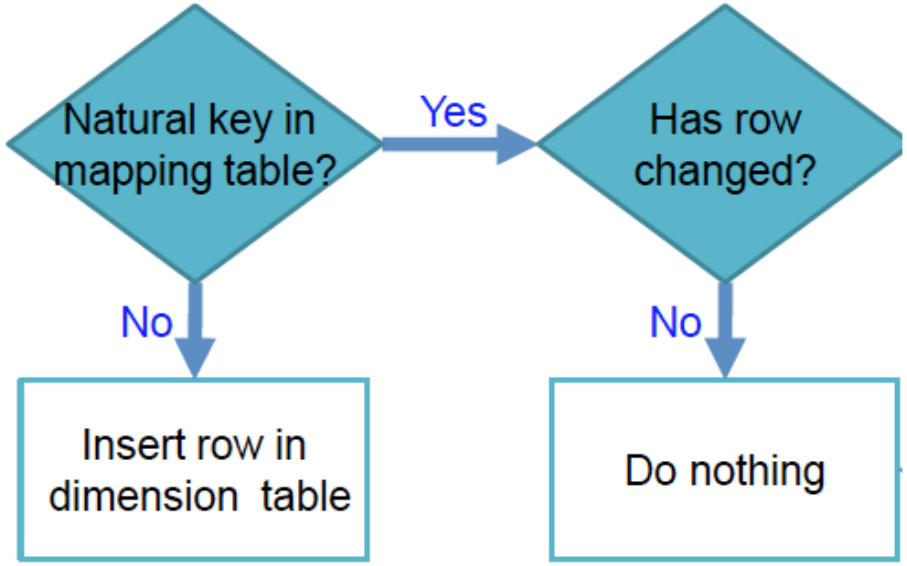
Loading of dimensions



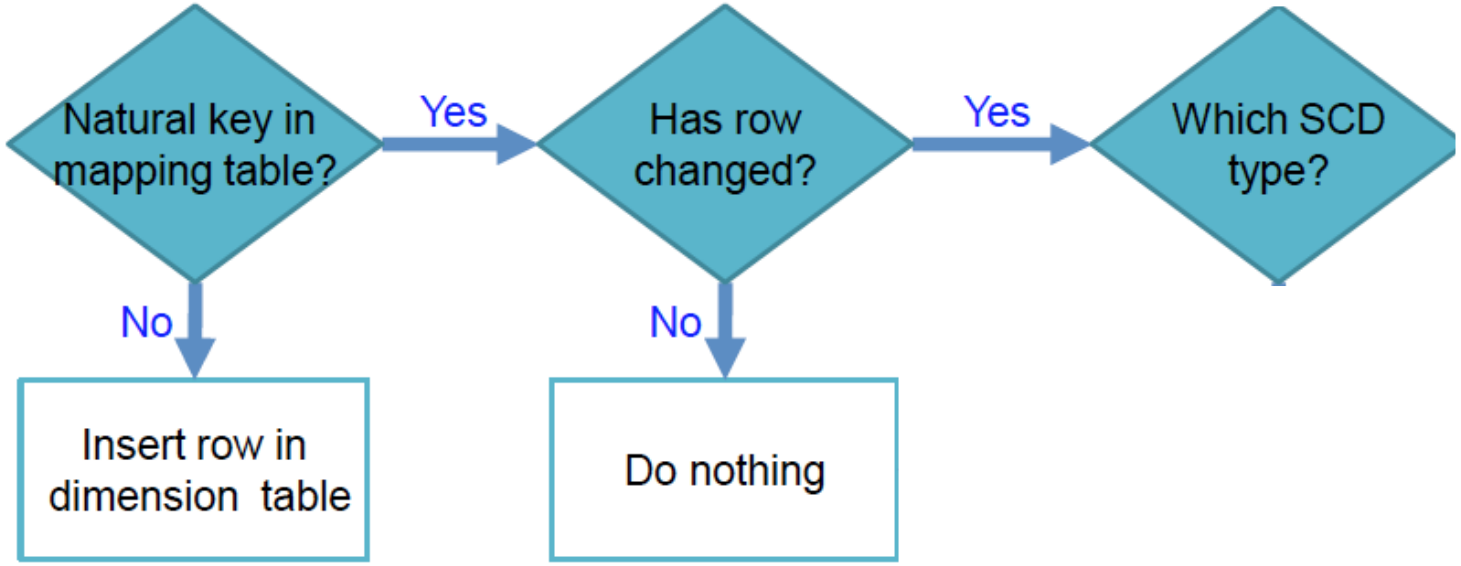
Loading of dimensions



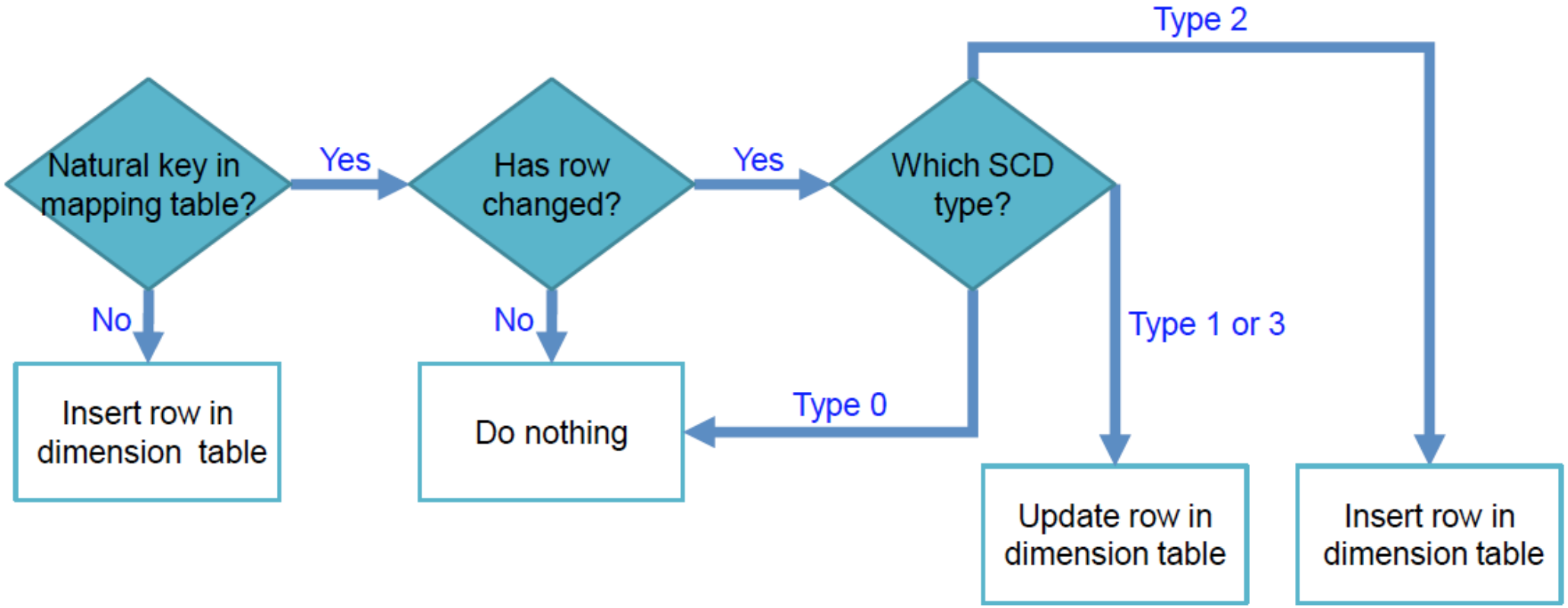
Loading of dimensions



Loading of dimensions



Loading of dimensions



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DIMENSIONAL MODELING

Loading of a fact

Loading a facttable

What’s the process of loading a fact?

Loading a facttable

Load the dimensions
If a dimensionvalue in unknown

- Don’t load the Record
- Use the Dummy record of the dimension

Load facttable with lookups to the fact

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DIMENSIONAL MODELING

Facttypes

Different types of facts

Fact table contains measures, metrics and other quantifiable information

Different types of facts

Three types of facts:

- Additive: can be used with aggregations like sum, average. Like Quantity, revenu, amounts..

- Semi-Additive: summable metrics for some combinations of dimensions like balance amounts. These are additive except time

- Non-Additive: these facts cannot be summed up for any of the dimensions like percentages. Not a best practice to store them in the dwh

Different types of facttables

- Transactional: 1 row is a measure at a point in time. The most atomic transaction is the combination of all the dimensions. Pitfall: a record exist when the event really occured.

- Periodic snapshot: summarizes data for a given period. Day, week, month. Not to a transactions. These tables contains many types of facts as many types of facts are allowed with the fact table grain.

- Accumulating snapshot: this facttable is used to show processevolutions. For example claimprocessing. There is a start point, intermediate steps and an end point. Specific for this fact table are the updates. There's only 1 row for a claim. If the status is changed then there won't be a new record created but the old one will be updated

- Factless: facts that are not numeric. For example attendence at events. This type of fact can also be used to analyze what didn't happen. For example a table with all the posibilities and 1 with everything that really happened

- Aggregate: simple rollups of atomic facts. For example per month - year

- Consolidated: combination of different types. for example Sales actuals and forecasts

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DIMENSIONAL MODELING

Exercise

Historical data at the Thomas More Airport

At the Thomas More Airport, we have a lot of various data. Off course, not everywhere we need to track the history. In this case we are going to store some of the changed data. How do you handle this historical changes? Be carefull, do not lose data!
Use as a startingpoint your last model of sales in the airport.

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DIMENSIONAL MODELING

Exercise call center

Case: Create a reporting model for our call center

We are called by a call center. The call center is searching for a self service reporting environment because they have different call centers, clients, different tools and different reporting needs. After our first review session we will start will the follow-up of call handling.
The model should be able to report per client the following information.

- Customer
- SLA reporting
- Call types
- Call center information
- Products related to a call

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DIMENSIONAL MODELING

Exercise The Warehouse

The warehouse

For the company ‘Thomas More warehouse’ a new datamodel needs to be created.
One of their main KPI’s is lead time.
Lead time is the time from the order until this is shipped towards the customer.

Besides the leadtime is the calculation of the following topics important:

- Cost of the products in stock
- How long are my products in stock
- Amount and cost price of the goods per buyer per warehouse
- Which store ordered wich product
- Are the ordered product shipped

The warehouse

Inbound

- Received number of pallets per pallettype, warehouse, day
- Received containers in TEU (Twenty Foot Equivalent Unit)
- Aggregation to week, month and year

The warehouse

Picking

- Order lines per warehouse, day
- Pickactions per warehouse, day
- Full Pallet picks per warehouse, day
- Aggregation to week, month, year

The warehouse

Outbound

- Number of pallet per palletype, warehouse, day
- Aggregation to week, month, year

The warehouse

Stock

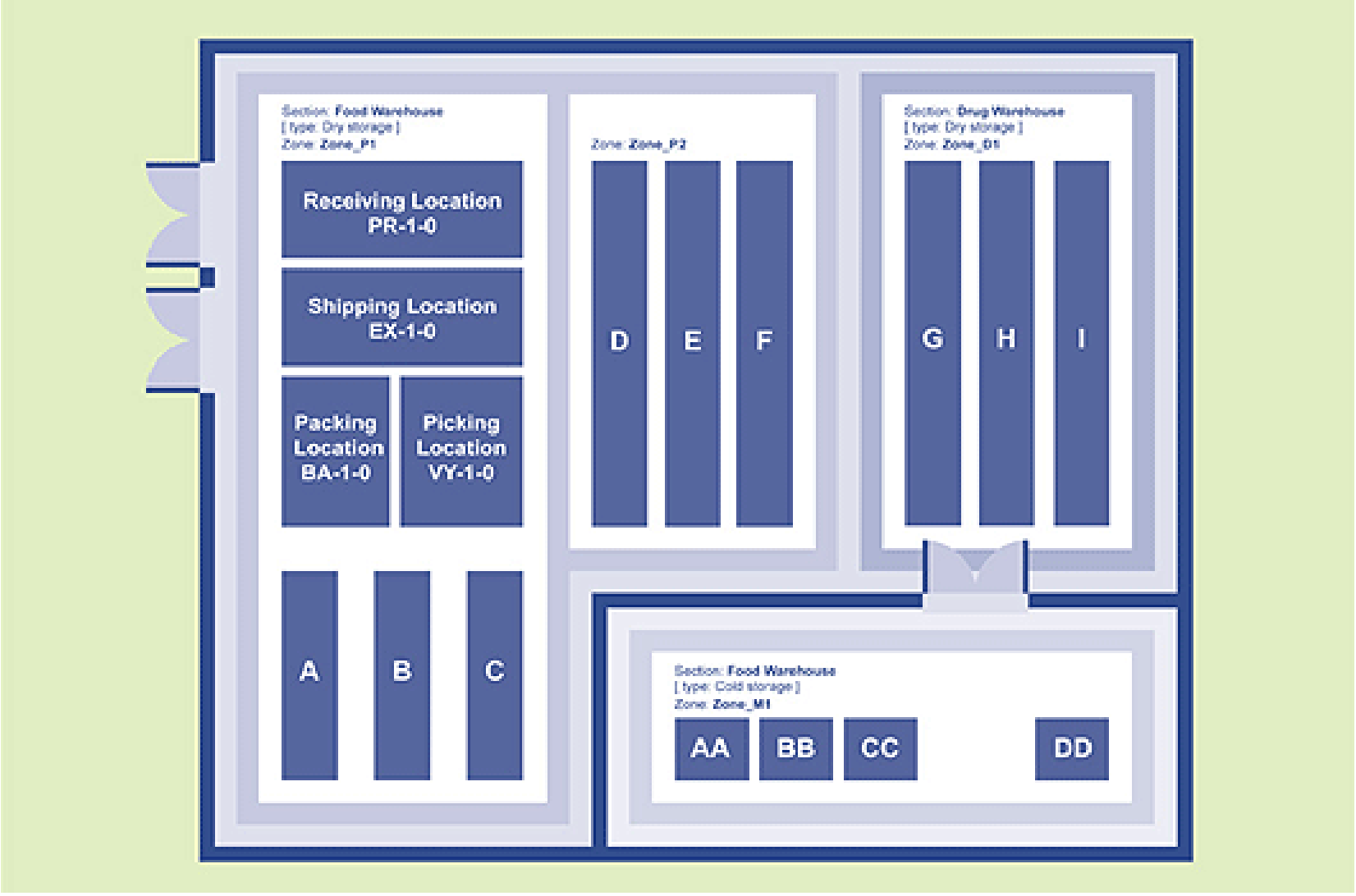
- Stock per pallettype, warehouse, day
- Stock value per warehouse, week in buying value
- Stock value per warehouse, week in salesvalue
- Drilldown to a lower relevant level like articlegroup, day
- Aggregation to week, month, year

The warehouse

Pay attention to slowly changing dimensions

What’s the lowest level of detail

The warehouse



The warehouse

Use the 4 steps to define the model
Deliver the following:

- Proces
- Spider
- Bus Matrix
- Physical data model